# StEOP Mathematical Foundations of Data Science

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**Abstract**

These are notes for the winter semester 2026 course. They are based on rough write-ups of the material covered in the lectures. These notes would not have been possible without the help of Mariia Mykhalova and Oleksandra Sierikova. Any remaining mistakes are my responsibility.

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# 1 Sets

## 1 Sets and membership

**Definition 1.1.** A *set* is a collection of objects. The objects belonging to the set are called its *elements*. We write

$$x \in A$$

when $x$ is an element of $A$, and $x \notin A$ otherwise.

Examples:

$$S = \{1, 2, 3\}, \qquad A = \{\text{apple}, 2, \text{student}\}, \qquad B = \{2, 3, 1\}.$$

The order in which elements are written does not matter, and repetitions do not create new elements. Thus

$$\{1, 2, 3, 3\} = \{3, 2, 1\}.$$

**Definition 1.2.** The *empty set* is the set with no elements. It is denoted by

$$\varnothing \qquad \text{or} \qquad \{\}.$$

In particular, $\varnothing \neq \{\varnothing\}$: the first set has no elements, whereas the second has one element.

**Definition 1.3.** The *cardinality* of a finite set $A$, denoted by $|A|$, is the number of elements in $A$.

Some sets you may be familiar already with are

$$\mathbb{N} = \{1, 2, 3, \ldots\}, \qquad \mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}, \qquad \mathbb{R} = (-\infty, \infty).$$

## 2 Equality and set-builder notation

**Definition 1.4.** Two sets $A$ and $B$ are equal if every element of $A$ is an element of $B$, and every element of $B$ is an element of $A$

A set can be described by a rule (set-builder notation):

$$A = \{a \in \mathbb{N} \,:\, a < 5\} = \{1, 2, 3, 4\}.$$

For example, the rational numbers can be written as

$$\mathbb{Q} = \left\{ \frac{m}{n} \,:\, m \in \mathbb{Z},\ n \in \mathbb{N} \right\}.$$

## 3 Ordered pairs and Cartesian products

An ordered pair $(a, b)$ records both the entries and their order. In general,

$$(a, b) \neq (b, a),$$

although $(a, a) = (a, a)$.

**Definition 1.5.** For sets $A$ and $B$, the *Cartesian product* is

$$A \times B = \{(a, b) \,:\, a \in A,\ b \in B\}.$$

For instance, if $A = \{1, 2\}$ and $B = \{a, b, c\}$, then

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

You may also know that

$$\mathbb{R} \times \mathbb{R} = \mathbb{R}^2,$$

where the latter is often identified with the Cartesian plane.

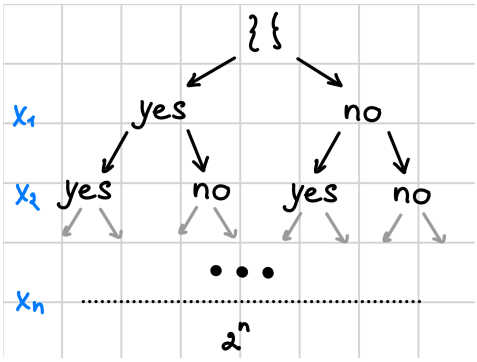
## 4 Subsets and power sets

**Definition 1.6.** A set $A$ is a *subset* of $B$, written $A \subset B$, if every element of $A$ belongs to $B$.
**Alternative notation.** $A \subseteq B$.

For $B = \{1, 2, 3\}$, examples of subsets are

$$\varnothing, \quad \{1\}, \quad \{2\}, \quad \{3\} \quad \{1, 2\}, \quad \{1, 3\}, \quad \{2, 3\}, \quad B.$$

**Theorem 1.7.** *If a finite set $S$ has $n$ elements, then it has $2^n$ subsets.*




*Proof.* Write $S = \{x_1, \ldots, x_n\}$. For each element $x_i$, there are two independent choices: include it in a subset, or do not include it. Hence the number of subsets is

$$\underbrace{2 \cdot 2 \cdots 2}_{n \text{ factors}} = 2^n.$$

Equivalently, subsets correspond to binary strings of length $n$. ∎

This is not a good proof yet; soon you will learn how make it more rigorous.

**Exercise 1.8.** What is true and what is not?

1. $1 \in \{1, \{1\}\}$
2. $1 \notin \{1, \{1\}\}$
3. $\{1\} \in \{1, \{1\}\}$

4. $\{1\} \subset \{1, \{1\}\}$

**Definition 1.9.** The *power set* of $A$ is the set of all subsets of $A$:

$$\mathcal{P}(A) = \{B \,:\, B \subset A\}$$

**Alternative notation.** Instead of $\mathcal{P}(A)$, we can also write $2^A$. The reason for the last notation is because we know that for finite $A$,

$$|\mathcal{P}(A)| = 2^{|A|} = |2^A|.$$

## 5 Basic set operations

**Definition 1.10.** The *union* and *intersection* of $A$ and $B$ are

$$A \cup B = \{x \,:\, x \in A \text{ or } x \in B\},$$

$$A \cap B = \{x \,:\, x \in A \text{ and } x \in B\}.$$

The word "or" is inclusive.

**Definition 1.11.** The *difference* of $A$ and $B$ is

$$A \setminus B = \{x \,:\, x \in A \text{ and } x \notin B\}.$$

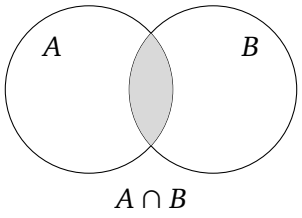
For

$$A = \{1, 2, 3, 4\}, \qquad B = \{2, 5\},$$

we have

$$A \cup B = \{1, 2, 3, 4, 5\}, \qquad A \cap B = \{2\},$$
$$A \setminus B = \{1, 3, 4\}, \qquad B \setminus A = \{5\}.$$

Often, it is convenient to use Euler-Venn diagram to denote operations on sets.




$A \cap B$

For example, with the Euler-Venn diagram you may quickly persuade yourself that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

You can persuade yourself but it won't constitute a proof. The proof must be formalized.

*Proof.* We must show that (i) $A \cap (B \cup C) \subset (A \cap B) \cup (A \cap C)$ and (ii) $(A \cap B) \cup (A \cap C) \subset A \cap (B \cup C)$.

(i) Let $x \in A \cap (B \cup C)$. This means that $x \in A$ and $x \in B \cup C$, which implies $x \in A$ and $x \in B$ or $x \in C$. If $x \in B$, then we have that $x \in A \cap B$ and hence, $x \in (A \cap B) \cup (A \cap C)$. The case when $x \in C$ is similar.

(ii) Let $x \in (A \cap B) \cup (A \cap C)$. This means that $x \in A \cap B$ or $x \in A \cap C$. Suppose first that $x \in A \cap B$. Then $x \in A$ and $x \in B$. Therefore, $x \in A$ and $x \in B \cup C$ and hence, $x \in A \cap (B \cup C)$. The case when $x \in A \cap C$ is similar. ∎

**Definition 1.12.** Let $U$ be a universal set. Think of it like an environment, where smaller sets live. The *complement* of $A$ is

$$A^c = U \setminus A.$$

**Alternative notation.** We may also denote it as $\bar{A}$ or $A'$.

For $U = \mathbb{N}$, the complement of the even numbers is the set of odd natural numbers.

## 6 Families of sets and sum notation

For sets $A_1, \ldots, A_n$, we use the notation

$$\bigcup_{i=1}^{n} A_i = A_1 \cup \cdots \cup A_n, \qquad \bigcap_{i=1}^{n} A_i = A_1 \cap \cdots \cap A_n,$$

and

$$\prod_{i=1}^{n} A_i = A_1 \times \cdots \times A_n.$$

If $A = \{a_1, a_2, \ldots, a_n\}$, we can write

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + \ldots + a_n, \qquad \prod_{i=1}^{n} a_i = a_1 a_2 \ldots a_n$$

to denote the sum and the product of elements in $A$.

**Exercise 1.13.** Find

$$|\{x \in \mathbb{N} : |x| < 10\}| = ? \qquad \mathcal{P}(\{2, \{3, 4\}\}) = ?$$

# 2 Propositional Logic

## 1 Propositions and truth values

**Definition 2.1.** A *proposition* is a statement with a definite truth value: true or false, but not both. Propositions are denoted by letters such as $P, Q, R$.

$$\begin{aligned} \mathrm{T} &= \mathrm{True} = 1 \\ \mathrm{F} &= \mathrm{False} = 0 \end{aligned}$$

We identify true with 1 and false with 0.

Questions, commands, and exclamations — such as "Will I pass the exam?", "Listen carefully!", and "What nice weather!" — are not propositions.

A proposition may contain a variable. For example, let $P(x)$ denote "$x$ is a prime number". Then $P(x)$ is a proposition for any particular value of $x$. For example, it is true for $x = 2, 3, 5, 7, 11, \ldots$ and false for $x = 1, 4, 6, 8, 9, \ldots$.

## 2 Logical connectives

We combine propositions using logical connectives. The basic connectives are "and", "or", and "not". "Implication" and "equivalence" can be expressed using the three basic connectives.

| | | |
|---|---|---|
| And | $\wedge$ | Conjunction |
| Or | $\vee$ | Disjunction |
| Not | $\neg$ | Negation |
| Implies | $\Longrightarrow$ | Implication |
| Equivalent to | $\Longleftrightarrow$ | Equivalence |

**Alternative notation.** In some textbooks negation $\neg P$ is denoted as $\overline{P}$ or $\sim P$, in programming it is often denoted as $!P$.

### 2.1 Conjunction, disjunction, and negation

**Definition 2.2.** The conjunction $P \wedge Q$ is true exactly when both $P$ and $Q$ are true. The disjunction $P \vee Q$ is true when at least one of $P, Q$ is true.

| $P$ | $Q$ | $P \wedge Q$ | $P \vee Q$ |
|---|---|---|---|
| $T$ | $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ | $T$ |
| $F$ | $T$ | $F$ | $T$ |
| $F$ | $F$ | $F$ | $F$ |

In mathematics, "or" is inclusive. In everyday English, however, it can be exclusive, as in "Take it or leave it."

**Definition 2.3.** The negation $\neg P$ means "it is not true that $P$."

| $P$ | $\neg P$ |
|---|---|
| $T$ | $F$ |
| $F$ | $T$ |

**Example 2.4.** We can construct a truth table for any statement using these basic rules. We first decompose the statement into the basic connectives "and", "or", and "not", and then apply their definitions. For example, consider

$$\underset{2}{\neg}(\underset{1}{P \wedge Q}) \underset{4}{\vee} \underset{3}{\neg R}$$

| $P$ | $Q$ | $R$ | $P \wedge Q$ | $\neg(P \wedge Q)$ | $\neg R$ | $\neg(P \wedge Q) \vee \neg R$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 | 0 |

In a truth table, it does not matter whether we use $T, F$ or $0, 1$, provided that we use the notation consistently.

## 2.2 Laws of logic

The following logical laws hold:

| | | | |
|---|---|---|---|
| $\neg(\neg P)$ | $\equiv$ | $P$ | (double negation) |
| $P \wedge Q$ | $\equiv$ | $Q \wedge P$ | (commutative) |
| $P \vee Q$ | $\equiv$ | $Q \vee P$ | |
| $(P \wedge Q) \wedge R$ | $\equiv$ | $P \wedge (Q \wedge R)$ | (associative) |
| $(P \vee Q) \vee R$ | $\equiv$ | $P \vee (Q \vee R)$ | |
| $P \wedge (Q \vee R)$ | $\equiv$ | $(P \wedge Q) \vee (P \wedge R)$ | (distributive) |
| $P \vee (Q \wedge R)$ | $\equiv$ | $(P \vee Q) \wedge (P \vee R)$ | |
| $\neg(P \wedge Q)$ | $\equiv$ | $\neg P \vee \neg Q$ | (De Morgan's laws) |
| $\neg(P \vee Q)$ | $\equiv$ | $\neg P \wedge \neg Q$ | |

Some of these laws can be deduced from others. You do not need to remember their names, but you do need to remember the laws themselves.

**Example 2.5.** Simplify $\neg(P \vee (\neg P \wedge Q)) \vee Q$.

*Solution.* We have

$$\neg(P \vee (\neg P \wedge Q)) \vee Q \equiv \neg((P \vee \neg P) \wedge (P \vee Q)) \vee Q \equiv \neg(P \vee Q) \vee Q \equiv (\neg P \wedge \neg Q) \vee Q$$
$$\equiv (\neg P \vee Q) \wedge (\neg Q \vee Q) \equiv (\neg P \vee Q)$$

❓ Explain each transformation above.

◇

## 3 Conditional propositions

**Definition 2.6.** The conditional proposition $P \implies Q$ means "if $P$, then $Q$." It is false only when $P$ is true and $Q$ is false.

| $P$ | $Q$ | $P \implies Q$ |
|---|---|---|
| $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ |
| $F$ | $T$ | $T$ |
| $F$ | $F$ | $T$ |

☛ Implication $P \implies Q$ can be equivalently written as $\neg P \vee Q$. Why?

In the implication $P \implies Q$, $P$ is the *hypothesis* and $Q$ is the *conclusion*. Equivalent English formulations include:

- $P$ is sufficient for $Q$;
- $Q$ is necessary for $P$;
- $P$ only if $Q$;
- $Q$ whenever $P$;
- $Q$ follows from $P$.

Given the implication $P \implies Q$, we can form its

$$\begin{aligned} \text{converse} &: Q \implies P, \\ \text{contrapositive} &: \neg Q \implies \neg P, \\ \text{inverse} &: \neg P \implies \neg Q. \end{aligned}$$

☛ Only one of these three statements is equivalent to the original implication $P \implies Q$. Which one?

## 4 Biconditionals and precedence

**Definition 2.7.** The biconditional $P \iff Q$ means that $P$ and $Q$ are equivalent (they have the same truth value). Equivalently,

$$P \iff Q \equiv (P \implies Q) \wedge (Q \implies P).$$

It may be read as "$P$ if and only if $Q$", "$P$ is equivalent to $Q$", or "$P$ is a necessary and sufficient condition for $Q$."

| $P$ | $Q$ | $P \iff Q$ |
|---|---|---|
| $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ |
| $F$ | $T$ | $F$ |
| $F$ | $F$ | $T$ |

The order in which we process the logical connectives is

$$\neg \quad \wedge \quad \vee \quad \implies \quad \iff$$

**Example 2.8.**

$$A \vee (B \wedge \neg A) \equiv (A \vee B) \wedge (A \vee \neg A) \equiv A \vee B$$

**Definition 2.9.** A *bit* is a symbol that can take either of two values: 0 or 1. The word "bit" is short for "binary digit".

**Definition 2.10.** A *Boolean variable* is a variable that can take either of two values: true or false.

Thus, bits can represent Boolean variables if we identify true with 1 and false with 0.

# 5 Quantifiers

Let $X = \{x_1, x_2, \ldots\}$, and let $P(x)$ mean "$x$ has property $A$".

How can we express the proposition "All elements of $X$ have property $A$"? We could, of course, write

$$P(x_1) \wedge P(x_2) \wedge P(x_3) \ldots,$$

but this is long and inefficient (and is not even well-defined when $X$ is infinite).

**Definition 2.11.** The *universal quantifier* $\forall x \in X$ means "for every element of $X$".

We can now write the preceding statement as $\forall x \in X\, P(x)$.

How can we express the proposition "There is an element of $X$ that has property $A$"? Again, we could write

$$P(x_1) \vee P(x_2) \vee P(x_3) \ldots,$$

but it has the same drawbacks as before.

**Definition 2.12.** The *existential quantifier* $\exists x \in X$ means "there exists an element in $X$."

We can now write the preceding statement as $\exists x \in X\, P(x)$.

Negation changes quantifiers:

$$\neg\big(\forall x \in X\, P(x)\big) \iff \exists x \in X\, \neg P(x),$$
$$\neg\big(\exists x \in X\, P(x)\big) \iff \forall x \in X\, \neg P(x).$$

**Example 2.13.**

a) ¬("There is a black cat in Vienna") means "There are no black cats in Vienna."

b) ¬("All students passed the exam") means "At least one student did not pass the exam."

**Example 2.14.** The statement "Every real number is the cube of a real number" can be written as

$$\forall x \in \mathbb{R}\ \exists y \in \mathbb{R} : x = y^3$$

Its negation is

$$\exists x \in \mathbb{R}\ \forall y \in \mathbb{R} : x \neq y^3.$$

*Proof.*

$$\begin{aligned}&\neg(\forall x \in \mathbb{R} : \exists y \in \mathbb{R} : x = y^3) \iff\\ &\exists x \in \mathbb{R} : \neg(\exists y \in \mathbb{R} : x = y^3) \iff\\ &\exists x \in \mathbb{R} : \forall y \in \mathbb{R} : x \neq y^3\end{aligned}$$

◇

**Exercise 2.15.** For each of the following statements, write it in plain English, determine whether it is true or false, and then write its negation.

a) $\forall x \in \mathbb{R} : x^2 > 0$

b) $\forall x \in \mathbb{R} : \exists n \in \mathbb{N} : x^n \geqslant 0$

c) $\exists a \in \mathbb{R} : \forall x \in \mathbb{R} : a \cdot x = x$

d) $\forall n \in \mathbb{N} : \exists X \subset \mathbb{N} : |X| = n$

# 6 Logic and sets

For fixed sets $A, B$, put

$$P : \ x \in A, \qquad Q : \ x \in B.$$

Strictly speaking, $P$ and $Q$ depend on $x$; for simplicity, we suppress this dependence. The logical connectives then correspond to set operations:

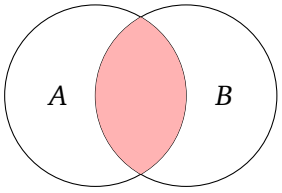
$$P \wedge Q \ \leftrightarrow \ x \in A \cap B, \qquad P \vee Q \ \leftrightarrow \ x \in A \cup B,$$

$$\neg P \ \leftrightarrow \ x \in A^c, \qquad P \implies Q \ \leftrightarrow \ A \subset B.$$

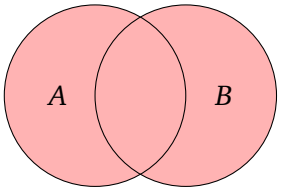
This correspondence explains the set-theoretic De Morgan's laws:

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

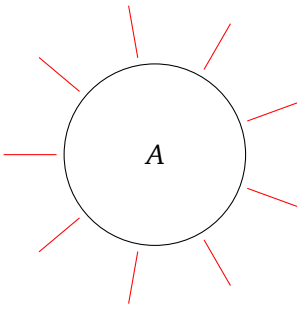
On the one hand, we can translate logical laws into set-theoretic laws. On the other hand, Euler–Venn diagrams can give us quick intuition about logical statements.



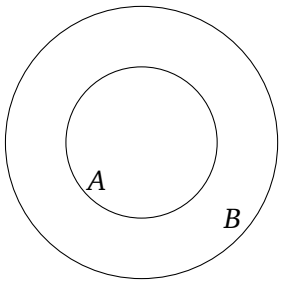

$P \wedge Q$



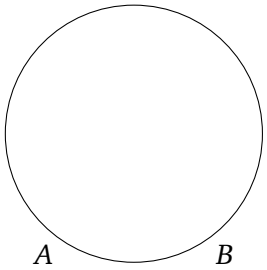

$P \vee Q$



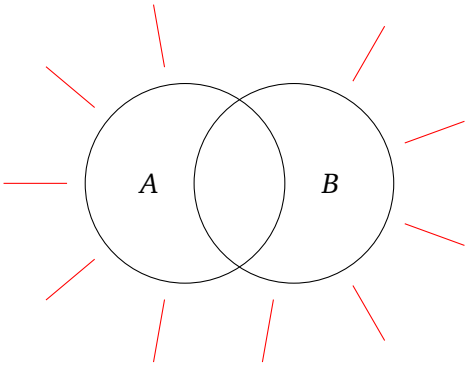

$\neg P$




$P \implies Q$




$P \iff Q$




De Morgan's law

**Exercise 2.16.** Prove

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

by translating membership into propositional logic.

**Exercise 2.17.** Show that

$$A = B \iff A \subset B \text{ and } B \subset A.$$

**Exercise 2.18.** Convert Example 2.5 into a set-theoretic statement.

**Exercise 2.19.** Prove the identity

$$A \cup B = A \cup (B \setminus A)$$

(i) directly by establishing both inclusions;

(ii) by using logical or set-theoretic laws.

**Exercise 2.20.** Prove that $\{4k + 5 : k \in \mathbb{Z}\} = \{4k + 1 : k \in \mathbb{Z}\}$.

# 3 Mathematical Induction

## 1 The induction principle

From afar, we can see a busy ice cream stand surrounded by people, both children and adults. We are told two facts about the queue for the ice cream stand:

1. The first person in the queue is a child.
2. If a child is followed by another person, that person is also a child.

These two facts imply that everyone in the queue is a child. This illustrates the principle of mathematical induction, one of the most powerful techniques for proving mathematical statements. Let us formalize it.

Suppose we want to prove a statement $P(n)$ for every integer $n \geqslant n_0$. We prove it in three steps:

1. **Base case:** prove $P(n_0)$.
2. **Induction hypothesis:** fix an arbitrary integer $n \geqslant n_0$ and assume that $P(n)$ is true.
3. **Induction step:** using the induction hypothesis, prove that $P(n+1)$ is true.

Alternatively, we can write it slightly shorter as

1. **Base case:** prove $P(n_0)$.
2. **Induction step:** prove $P(n) \implies P(n+1)$.

Persuade yourself that both ways are equivalent.

The best way to master induction is by solving problems.

**Example 3.1.** Prove that, for every $n \in \mathbb{N}$,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

*Proof.*

1. For the base case $n = 1$,
$$1^2 = \frac{1 \cdot 2 \cdot 3}{6}.$$
2. Assume that
$$1^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$
3. We must prove that
$$1^2 + \cdots + n^2 + (n+1)^2 = \frac{(n+1)(n+2)(2n+3)}{6}.$$

By the induction hypothesis,

$$\begin{aligned} 1^2 + \cdots + n^2 + (n+1)^2 &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 \\ &= (n+1)\left(\frac{n(2n+1)}{6} + n + 1\right) \\ &= \frac{(n+1)(2n^2+7n+6)}{6} \\ &= \frac{(n+1)(n+2)(2n+3)}{6}. \end{aligned}$$

◇

For integers $a$ and $b$, the notation $a \mid b$ means that $a$ divides $b$. Equivalently,

$$a \mid b \quad \Longleftrightarrow \quad \exists c \in \mathbb{Z} : \; b = ca.$$

**Example 3.2.** Prove that $6 \mid n^3 + 5n$ for every $n \in \mathbb{N}$.

*Proof.*

1. For the base case $n = 1$, we have $6 \mid 1^3 + 5$.
2. Assume that $6 \mid n^3 + 5n$.
3. We must prove that $6 \mid (n+1)^3 + 5(n+1)$. We have

$$(n+1)^3 + 5(n+1) = \underline{(n^3 + 5n)} + 3n^2 + 3n + \underline{6}.$$

The term $n^3 + 5n$ is divisible by 6 by the induction hypothesis, and $6 \mid 6$. It therefore remains to show that $6 \mid 3n^2 + 3n$. This follows from

$$3n^2 + 3n = 3n(n+1),$$

because the product of two consecutive integers is even.

◇

**Example 3.3.** Compute $S_n = 1 + 3 + 5 + \cdots + (2n-1)$.

*Proof.* Calculating the first few values gives

$$S_1 = 1, \qquad S_2 = 4, \qquad S_3 = 9, \qquad S_4 = 16,$$

which suggests the conjecture $S_n = n^2$. We prove it by induction.

1. For the base case, $S_1 = 1 = 1^2$.
2. Assume that $S_n = n^2$.
3. The next summand is $2n + 1$, so

$$S_{n+1} = S_n + (2n+1) = n^2 + 2n + 1 = (n+1)^2.$$

◇

**Example 3.4.** Prove that, for every $n \in \mathbb{N}$,

$$\frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{n^2} \leqslant 2.$$

Explain why it is impossible to prove this inequality directly by induction. Surprisingly, we shall prove a stronger inequality instead.

**Example 3.5.** Prove that, for every $n \in \mathbb{N}$,

$$\sum_{i=1}^{n} \frac{1}{i^2} \leqslant 2 - \frac{1}{n}.$$

*Proof.*

1. For the base case $n = 1$,
$$1 = 2 - \frac{1}{1}.$$
2. Assume that
$$\sum_{i=1}^{n} \frac{1}{i^2} \leqslant 2 - \frac{1}{n}.$$
3. We must prove that
$$\sum_{i=1}^{n+1} \frac{1}{i^2} \leqslant 2 - \frac{1}{n+1}.$$
By the induction hypothesis,
$$\begin{aligned}\sum_{i=1}^{n+1} \frac{1}{i^2} &= \sum_{i=1}^{n} \frac{1}{i^2} + \frac{1}{(n+1)^2} \\ &\leqslant 2 - \frac{1}{n} + \frac{1}{(n+1)^2}.\end{aligned}$$
It remains to verify that
$$2 - \frac{1}{n} + \frac{1}{(n+1)^2} \leqslant 2 - \frac{1}{n+1}.$$
This is equivalent to
$$\frac{1}{(n+1)^2} \leqslant \frac{1}{n(n+1)},$$
which is true.

◇

**Lemma 3.6** (Triangle inequality for two numbers)**.** *For all real numbers $x$ and $y$,*

$$|x + y| \leqslant |x| + |y|.$$

*Proof.* Since both sides are nonnegative, this inequality is equivalent to

$$|x + y|^2 \leqslant (|x| + |y|)^2 \quad \Longleftrightarrow \quad x^2 + y^2 + 2xy \leqslant x^2 + y^2 + 2|x||y|,$$

which is true. ■

**Theorem 3.7** (Triangle inequality)**.** *For real numbers $x_1, \dots, x_n$,*

$$|x_1 + \cdots + x_n| \leqslant |x_1| + \cdots + |x_n|.$$

*Proof.*

1. For the base case $n = 1$, the statement is the equality
$$|x_1| = |x_1|.$$
2. Assume that
$$\left|\sum_{i=1}^{n} x_i\right| \leqslant \sum_{i=1}^{n} |x_i|.$$
3. We must prove the statement for $n + 1$. We have
$$\begin{aligned}
\left|\sum_{i=1}^{n+1} x_i\right| &= \left|\sum_{i=1}^{n} x_i + x_{n+1}\right| && \text{(by Lemma 3.6)} \\
&\leqslant \left|\sum_{i=1}^{n} x_i\right| + |x_{n+1}| && \text{(by induction hypothesis)} \\
&\leqslant \sum_{i=1}^{n} |x_i| + |x_{n+1}| \\
&= \sum_{i=1}^{n+1} |x_i|.
\end{aligned}$$

■

**Definition 3.8.** For $n \in \mathbb{N}$, define *n-factorial* as

$$n! = 1 \cdot 2 \cdots n,$$

and set $0! = 1$.

**Exercise 3.9.** For $n \in \mathbb{N}$, prove that

$$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n + 1)! - 1.$$

**Exercise 3.10.** A $16 \times 16$ chessboard has one square removed. Prove that, for every integer $n \geqslant 1$, the remaining board can be tiled by L-shaped trominoes (you can rotate them arbitrary).

This is a fairly non-trivial problem. Instead of thinking how to solve this particular problem, let's generalize it.

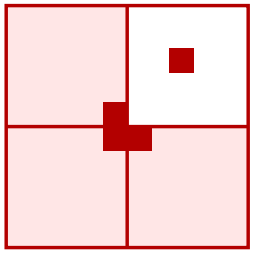
**Example 3.11.** A $2^n \times 2^n$ chessboard has one square removed. Prove that, for every integer $n \geqslant 1$, the remaining board can be tiled by L-shaped trominoes.



tromino

*Proof.*

1. For the base case $n = 1$, a $2 \times 2$ board with one square removed is exactly one L-shaped tromino.
2. Assume that the statement is true for some integer $n \geqslant 1$.
3. Consider a $2^{n+1} \times 2^{n+1}$ board with one square removed. Divide it into four $2^n \times 2^n$ quadrants. Exactly one quadrant contains the missing square. Place one L-shaped tromino at the centre so that it covers one central square in each of the other three quadrants. Each quadrant now has exactly one unavailable square, so the induction hypothesis applies to all four quadrants.



◇

## 2 Strong induction

Ordinary induction assumes $P(n)$ in order to prove $P(n+1)$. Sometimes the induction step requires several earlier cases. In strong induction, we assume that $P(k)$ holds for every integer $k$ from the initial value up to $n$, and we use these assumptions to prove $P(n+1)$. The next example illustrates this form of induction.

**Example 3.12.** Prove that every integer $n \geqslant 8$ can be expressed using only 3-euro and 5-euro coins. Equivalently, prove that

$$n = 3a + 5b$$

for some $a, b \in \mathbb{N} \cup \{0\}$.

*Proof.*

1. The base cases are

$$8 = 3 \cdot 1 + 5 \cdot 1, \qquad 9 = 3 \cdot 3 + 5 \cdot 0, \qquad 10 = 3 \cdot 0 + 5 \cdot 2.$$

2. Let $n \geqslant 10$, and assume that every integer $k$ with $8 \leqslant k \leqslant n$ can be written as $k = 3a + 5b$, where $a, b \in \mathbb{N} \cup \{0\}$.
3. We show that $n+1$ can also be written in this form. Since $n + 1 \geqslant 11$, we have $8 \leqslant (n+1) - 3 \leqslant n$. The induction hypothesis therefore applies to $(n+1) - 3 = n - 2$: there exist $a', b' \in \mathbb{N} \cup \{0\}$ such that

$$n - 2 = 3a' + 5b'.$$

Adding 3 to both sides gives

$$n = 3(a' + 1) + 5b'.$$

This is the required form, with $a = a' + 1$ and $b = b'$; both are nonnegative integers.

◇

**Exercise 3.13.** Let

$$f_1 = f_2 = 1, \qquad f_{n+1} = f_n + f_{n-1}.$$

Prove that $f_n \leqslant 2^{n-1}$.

# 4 Proofs

In mathematics, statements are established through proof. Often, we begin with a statement $P$ that is known to be true and use it to prove another statement $Q$. In other words, starting from $P$, we derive $Q$.

## 1 Direct proofs

One way of proving $Q$ is to show *directly* that the implication $P \implies Q$ is true. Consider the following problem.

**Example 4.1.** The square of an odd integer is odd.

*Proof.* If $x$ is odd, then $x = 2n + 1$ for some $n \in \mathbb{Z}$. Hence

$$x^2 = (2n+1)^2 = 4n^2 + 4n + 1 = 2\big(2n(n+1)\big) + 1,$$

which is also an odd number. ◇

This was an example of a *direct proof*.

**Example 4.2.** If $a \mid b$ and $b \mid c$, then $a \mid c$.

*Proof.* There are integers $m, n$ such that $b = am$ and $c = bn$. Thus

$$c = (am)n = a(mn),$$

so $a \mid c$. ◇

**Example 4.3.** If $x$ is even, then $x^2 - 6x + 5$ is odd.

*Proof.* Write $x = 2n$. Then

$$x^2 - 6x + 5 = 4n^2 - 12n + 5 = 2(2n^2 - 6n + 2) + 1,$$

which is odd. ◇

## 2 Proof by contraposition

The implication

$$P \implies Q$$

is equivalent to its contrapositive

$$\neg Q \implies \neg P.$$

Sometimes, it is the second implication that is easier to prove.

**Example 4.4.** If $3n + 2$ is odd, then $n$ is odd.

*Proof.* We prove the contrapositive: if $n$ is even, then $3n + 2$ must be even. Since $n$ is even, we can write $n = 2k$ for $k \in \mathbb{Z}$. Then $3n + 2 = 6k + 2 = 2(3k + 1)$, which is even. ◇

**Example 4.5.** If $n^2$ is even, then $n$ is even.

*Proof.* The contrapositive says that if $n$ is odd, then $n^2$ is odd, which was proved in Example 4.1. ◇

## 3 Proof by contradiction

To prove a statement $Q$ by contradiction, assume $\neg Q$ and derive an impossibility, that is $P \wedge \neg Q$ must be false.

**Example 4.6.** If the sum of the positive divisors of $n > 1$ is $n + 1$, then $n$ is prime.

*Proof.* Assume $n$ is not prime. Then $n$ has a positive divisor $a$ with $1 < a < n$. The positive divisors include at least $1, a, n$, whose sum is strictly greater than $n + 1$, a contradiction. ◇

**Example 4.7.** The number $\sqrt{2}$ is irrational.

*Proof.* Assume

$$\sqrt{2} = \frac{m}{n}$$

with $m, n \in \mathbb{N}$ having no common factor. Then $m^2 = 2n^2$, so $m$ is even: $m = 2k$. Substitution gives

$$2n^2 = 4k^2, \qquad n^2 = 2k^2,$$

so $n$ is also even. Thus $m, n$ have the common factor 2, contradicting the choice of the fraction. ◇

**Theorem 4.8** (Euclid)**.** *There are infinitely many prime numbers.*

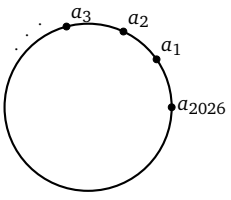
*Proof.* Assume that $p_1, \ldots, p_r$ are all the primes. Consider

$$N = p_1 p_2 \cdots p_r + 1.$$

None of $p_1, \ldots, p_r$ divides $N$, because division leaves remainder 1. But every integer greater than 1 has a prime divisor. That divisor is not on the list, a contradiction. ■

**Example 4.9.** Consider a cyclic sequence $a_1, \ldots, a_{2026}$ satisfying

$$a_i = \frac{a_{i-1} + a_{i+1}}{2} \quad \text{where } a_0 = a_{2026}.$$




Prove that all numbers $a_i$ are equal.

*Proof.* Suppose that not all elements are equal. Let $M$ be the maximum element of the sequence. Because the sequence is cyclic and not constant, there is an index $i$ such that

$$a_i = M \qquad \text{and} \qquad a_{i-1} \neq a_i.$$

This implies that $a_{i-1} < a_i$. On the other hand,

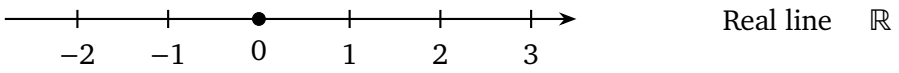
$$a_i = \frac{1}{2}(a_{i-1} + a_{i+1}) \leqslant \frac{1}{2}(a_{i-1} + a_i) \qquad \Longrightarrow \qquad a_i \leqslant a_{i-1}.$$

But this contradicts our choice of $(a_{i-1}, a_i)$. Hence all numbers must be equal. ◇

# 5 Numbers

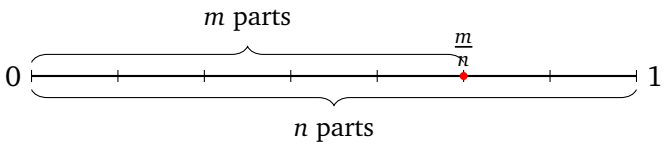
## 1 The standard number systems

**Numbers**




$$\mathbb{N} = \{1, 2, 3, \ldots\}, \qquad \mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\},$$
$$\mathbb{Q} = \left\{\frac{m}{n} \,:\, m \in \mathbb{Z},\ n \in \mathbb{N}\right\}, \qquad \mathbb{R} = \text{the real numbers.}$$

A rational number $\frac{m}{n}$ with $m \leqslant n$ for $m, n \in \mathbb{N}$ is defined exactly as the figure below shows. Other cases are a trivial extension.




There are different ways to write the same fraction:

$$\frac{1}{2} = \frac{4}{8} = \frac{16}{32}$$

**Definition 5.1.** A fraction $\frac{m}{n}$ is *irreducible*, or in lowest terms, if $m$ and $n$ have no common factors greater than 1.

## 2 Irrational numbers

**Theorem 5.2.** *Between any two distinct rational numbers there is another rational number.*

*Proof.* If $a, b \in \mathbb{Q}$ and $a < b$, then their midpoint $c = \frac{a+b}{2}$ is rational and satisfies $a < c < b$. Indeed, if $a = \frac{m}{n}$, $b = \frac{p}{q}$, then

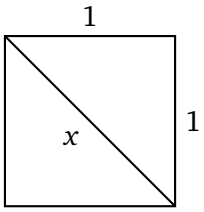
$$c = \frac{mq + pn}{2nq} \in \mathbb{Q}.$$

■

**Question:** Are there other numbers in $\mathbb{R}$ besides $\mathbb{Q}$?

**Proposition 5.3.** *There is a real number $x$ such that $x^2 = 2$.*

*Proof.*




By Pythagoras

$$1^2 + 1^2 = x^2$$
$$2 = x^2$$

■

**Notation:** $x = \sqrt{2}$

**Theorem 5.4.** $\sqrt{2} \notin \mathbb{Q}$ *(Proved last lecture)*

**Definition 5.5.** The set $\mathbb{R} \setminus \mathbb{Q}$ is called the set of *irrational* numbers.

- Can the sum of two irrational numbers be rational? Yes:

$$\sqrt{2} + (-\sqrt{2}) = 0.$$

- Can the sum or rational number and irrational be rational? No. If $a \in \mathbb{Q}$ and $b \in \mathbb{R} \setminus \mathbb{Q}$, and $a + b \in \mathbb{Q}$, then $(a + b) - a \in \mathbb{Q}$ as a difference of two rational numbers.
- Can the product of a nonzero rational number and an irrational number be rational? No. If $a \neq 0$, $a \in \mathbb{Q}$, and $ab \in \mathbb{Q}$, then $b = (ab)/a \in \mathbb{Q}$.

**Theorem 5.6.** *Between any two distinct real numbers there is an irrational number.*

*Proof.* Let $a < b$. Choose $n \in \mathbb{N}$ large enough that

$$\frac{\sqrt{2}}{n} < b - a.$$

If $a \in \mathbb{Q}$, then $a + \sqrt{2}/n$ is irrational and lies between $a$ and $b$. If $a \notin \mathbb{Q}$, choose $a + 1/n$, which remains irrational for a suitable $n$ and lies between $a$ and $b$. ■

## 3 Decimal representations

The standard decimal notation of integers that you commonly use reads as follows:

$$124 = 1 \cdot 10^2 + 2 \cdot 10^1 + 4 \cdot 10^0.$$

In general, the notation $m = (b_k b_{k-1} \dots b_1 b_0)_{10}$, where $b_i \in \{0, 1, \dots, 9\}$, means that

$$m = \sum_{i=0}^{k} b_i \cdot 10^i.$$

In practice we usually don't write $()_{10}$ unless there is a confusion. This is called a decimal representation of integers. Now, consider a more general case of real numbers.

$$\left.\begin{aligned} \pi &= 3.141\ldots \\ \frac{10}{9} &= 1.111\ldots \\ \frac{2}{5} &= 0.400\ldots \end{aligned}\right\} \text{ decimal representation}$$

A decimal is written as

$$x = b_k b_{k-1} \cdots b_0 . a_1 a_2 a_3 \cdots ,$$

where each digit $a_i$ and $b_i$ belongs to $\{0, 1, \ldots, 9\}$. Its value is

$$x = \sum_{i=0}^{k} b_i 10^i + \sum_{i=1}^{\infty} a_i 10^{-i}.$$

**Exercise 5.7.** Verify that the above definition is indeed valid for $\frac{10}{9}$. For this, you may want to use the formula

$$q + q^2 + q^3 + \cdots = \frac{q}{1-q} \qquad \text{for } 0 < q < 1.$$

Also prove this formula.

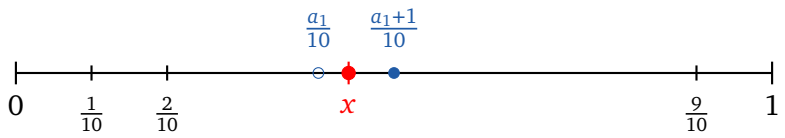
**Theorem 5.8.** *Every $x \in (0, 1)$ has a decimal representation.*

Alternatively, this means that we can approximate every $x \in (0, 1)$ by a decimal number with arbitrary accuracy.

*Proof.* Let $x \in (0, 1)$. We construct the decimal digits $a_1, a_2, a_3, \cdots \in \{0, 1, \ldots, 9\}$ inductively.

**Step 1: Determining the first digit $a_1$.** Partition the interval $(0, 1]$ into 10 subintervals of width $\frac{1}{10}$:

$$\left(\frac{0}{10}, \frac{1}{10}\right], \left(\frac{1}{10}, \frac{2}{10}\right], \ldots, \left(\frac{9}{10}, \frac{10}{10}\right]$$




Since these subintervals partition $(0, 1]$, there exists a unique integer $a_1 \in \{0, 1, \ldots, 9\}$ such that:

$$\frac{a_1}{10} < x \leqslant \frac{a_1 + 1}{10}$$

Thus, the first decimal digit of $x$ is $a_1$, giving the first approximation $x \approx 0.a_1$.

**Step 2: Constructing the $n$-th digit $a_n$.** Now suppose that $a_1, a_2, \ldots, a_{n-1}$ are already chosen. We zoom into the remaining subinterval of size $\frac{1}{10^{n-1}}$ and divide it into 10 smaller slices of size $\frac{1}{10^n}$. Then the number

$$x - \sum_{k=1}^{n-1} \frac{a_k}{10^k}$$

must fall into one of these intervals of size $\frac{1}{10^n}$. This means that there exists a unique integer $a_n$ satisfying:

$$\frac{a_n}{10^n} < x - \sum_{k=1}^{n-1} \frac{a_k}{10^k} \leqslant \frac{a_n + 1}{10^n}.$$

But this is the same as

$$\sum_{i=1}^{n} \frac{a_k}{10^k} < x \leqslant \sum_{i=1}^{n-1} \frac{a_k}{10^k} + \frac{a_n + 1}{10^n}.$$

Since with such a construction we can reach any $n \in \mathbb{N}$, we get a valid decimal expansion.

■

## 3.1 Non-uniqueness

Decimal representations are not always unique:

$$0.5000\ldots = 0.4999\ldots, \qquad 1.0000\ldots = 0.9999\ldots.$$

More generally, a terminating decimal has a second representation ending in repeating 9's:

$$0.26000\ldots = 0.25999\ldots.$$

**Theorem 5.9.** *If two decimal representations of the same real number are different, then for some $k$ they agree up to the $(k-1)$-st digit, and from the $k$-th digit onward they are of the form*

$$\cdots a_k 999\ldots \qquad \textit{and} \qquad \cdots (a_k + 1)000\ldots$$

*Proof.* Let $x$ be our number. It is sufficient to show this for $x \in (0, 1]$. Let $k$ be the first time where two representations of $x$ start to differ:

$$\begin{aligned} x &= 0.a_1 a_2 \cdots a_{k-1} a_k \ldots \\ &= 0.a_1 a_2 \cdots a_{k-1} b_k \ldots \end{aligned}$$

Without loss of generality, assume that $a_k < b_k$. The two equations above imply respectively that

$$\begin{aligned} x &\leqslant 0.a_1 a_2 \cdots a_{k-1}(a_k + 1)00\ldots \\ x &\geqslant 0.a_1 a_2 \cdots a_{k-1} a_k 99\ldots \end{aligned}$$

But this is possible only when

$$0.a_1 a_2 \cdots a_{k-1} a_k 99\ldots = 0.a_1 a_2 \cdots a_{k-1}(a_k + 1)00\ldots$$

Hence, the proof is complete. ■

## 3.2 Periodic decimals

**Definition 5.10.** A decimal is *periodic* if, after some finite initial block, a finite block of digits repeats forever. For example,

$$0.\overline{3} = 0.333\ldots, \qquad 0.12\overline{34} = 0.12343434\ldots.$$

**Theorem 5.11.** *Every rational number has an eventually periodic decimal representation.*

The best way to "feel" the proof is to perform a long division of concrete numbers, say $\frac{1}{7}$.

*Proof.* Let $m, n \in \mathbb{N}$ and consider $q = \frac{m}{n} \in \mathbb{Q}$. Without loss of generality assume that $m < n$. When you divide $m$ over $n$ using long-division algorithm, each step produces a remainder from the finite set $\{0, 1 \ldots, n-1\}$. Hence either a remainder becomes 0, in which case the decimal terminates, or some remainder repeats. Once a remainder repeats, all following digits repeat periodically. ∎

**Theorem 5.12.** *Every periodic decimal represents a rational number.*

*Proof.* Let $x \in \mathbb{R}$ be a periodic decimal. This means that

$$x = d + 0.a_1 \cdots a_m \overline{b_1 \cdots b_k},$$

where $d \in \mathbb{Z}$ and $a_i, b_i \in \{0, 1, \ldots, 9\}$ for all $i$. Then

$$10^m x = d \cdot 10^m + (a_1 \cdots a_m) + 0.\overline{b_1 \cdots b_k}$$

We see that $x$ will be rational if we prove that the tail $t = 0.\overline{b_1 \cdots b_k}$ is rational. Let's prove this. We have that

$$10^k t = b_1 \cdots b_k + t.$$

Therefore,

$$t = \frac{b_1 \cdots b_k}{10^k - 1} \in \mathbb{Q}$$

and the proof is complete. ∎

## 4 Positional and non-positional systems

A numeral system may be *positional* or *non-positional*. Roman numerals are a non-positional example (almost):

$$2 = \text{II}, \qquad 25 = \text{XXV}, \qquad 2026 = \text{MMXXVI}.$$

In a positional system with base $b$, a finite digit string

$$a_n a_{n-1} \cdots a_1 a_0,$$

where

$$a_i \in \{0, 1, \ldots, b-1\},$$

represents

$$a_n b^n + a_{n-1} b^{n-1} + \cdots + a_1 b + a_0.$$

**Theorem 5.13.** *Let $b \in \mathbb{N}$, $b \geqslant 2$. Every $m \in \mathbb{N}$ has a unique base-$b$ representation*

$$m = a_k b^k + \cdots + a_1 b + a_0, \qquad 0 \leqslant a_i < b, \quad a_k \neq 0.$$

*Proof.* Try to prove it by yourself. ∎

## 4.1 Common bases

### Binary

In base 2, the digits are 0 and 1. For example,

$$(10011)_2 = 1 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2 + 1 = 19.$$

### Hexadecimal

Base 16 uses the digits

$$0, 1, \ldots, 9, A, B, C, D, E, F,$$

with $A = 10, \ldots, F = 15$. For example,

$$(A2)_{16} = 10 \cdot 16 + 2 = 162.$$

### Sexagesimal

**Question:** Is there another numeral system that is still relevant today?
**Answer:** Yes. We are still influenced by the sexagesimal system, which uses 60 as its base. It goes back to ancient Mesopotamia!

**Example 5.14.** Convert 45 from base 10 to base 2. We have

$$45 = 32 + 8 + 4 + 1,$$

hence

$$45 = (101101)_2.$$

**Exercise 5.15.** Convert the following numbers.

1. $(110101)_2$ to base 10;
2. 2025 to base 16;
3. $(3A7)_{16}$ to base 10.

# 6 Integers

## 1 Divisibility and the division algorithm

**Definition 6.1.** For $a, b \in \mathbb{Z}$, $a \neq 0$, we say that *a divides b*, written $a \mid b$, if $b = ac$ for some $c \in \mathbb{Z}$.

**Theorem 6.2.** *For $a \in \mathbb{Z}$ and $b \in \mathbb{N}$, there exist unique integers $q, r$ such that*

$$a = bq + r, \qquad 0 \leqslant r \leqslant b - 1.$$

*The integer $q$ is the quotient and $r$ is the remainder.*

*Proof.* Existence. Consider $\dfrac{a}{b} \in \mathbb{Q}$. There exists $q \in \mathbb{Z}$ such that

$$q \;\leqslant\; \frac{a}{b} \;<\; q + 1.$$

Then let $r = a - bq$. Since $q \leqslant \frac{a}{b}$, we have that $r = a - bq \geqslant 0$. It remains to prove that $r \leqslant b - 1$. From $\frac{a}{b} < q + 1$, it follows that

$$a < bq + b \qquad \Longleftrightarrow \qquad r = a - bq < b.$$

Uniqueness. Suppose $q, r$ are not unique, that is

$$a = bq + r = bq' + r', \qquad 0 \leqslant r < b, \quad 0 \leqslant r' < b.$$

Subtracting,

$$b(q - q') = r' - r.$$

Since $0 \leqslant r < b$ and $0 \leqslant r' < b$, we have $-b < r' - r < b$ so

$$|b(q - q')| = |r' - r| < b.$$

As $b > 0$, this forces $|q - q'| < 1$. Since $q - q' \in \mathbb{Z}$, the only integer with absolute value strictly less than 1 is 0. Hence $q = q'$, and then $r' - r = b(q - q') = 0$ gives $r = r'$. ∎

**Definition 6.3.** A number $p$ is called *prime* if it has just two divisors: 1 and $p$.

**Remark 6.4.** In particular, 1 is *not* a prime number, since 1 has only one divisor (itself), not two.

## 2 Common divisors and greatest common divisors

**Definition 6.5.** A positive integer $d$ is a *common divisor* of $a, b$ if $d \mid a$ and $d \mid b$. The greatest common divisor, denoted $\gcd(a, b)$, is the largest positive common divisor.

Examples: $\gcd(6, 10) = 2$, $\gcd(8, 16) = 8$, $\gcd(21, 9) = 3$, $\gcd(12, 35) = 1$.

**Lemma 6.6.** *If $a = bq + r$, then* $\gcd(a, b) = \gcd(b, r)$

*Proof.* Let $c \mid a$ and $c \mid b$. Then

$$c \mid a - bq \quad \Longrightarrow \quad c \mid r.$$

On the other hand, if $c \mid b$ and $c \mid r$, then

$$c \mid bq + r \quad \Longrightarrow \quad c \mid a.$$

In other words, we proved that the sets of common divisors of $a$ and $b$, and of $b$ and $r$, are the same. Hence $\gcd(a, b) = \gcd(b, r)$. ■

This result immediately leads to an algorithm for finding the greatest common divisor of two numbers.

EUCLID'S ALGORITHM

**Input:** $a, b \in \mathbb{N}$.
**Output:** $\gcd(a, b)$.

1. Divide $a$ by $b$:
$$a = q_1 b + r_1, \qquad 0 \leqslant r_1 < b.$$
If $r_1 = 0$, return $b$.
2. Continue dividing by the most recent non-zero remainder:
$$\begin{aligned} b &= q_2 r_1 + r_2, & 0 \leqslant r_2 < r_1, \\ r_1 &= q_3 r_2 + r_3, & 0 \leqslant r_3 < r_2, \\ &\vdots \\ r_{n-2} &= q_n r_{n-1} + r_n, & 0 \leqslant r_n < r_{n-1}, \\ r_{n-1} &= q_{n+1} r_n + 0. \end{aligned}$$
3. Return $r_n$, the last non-zero remainder.

**Theorem 6.7.** *The Euclid's algorithm outputs correctly* $\gcd(a, b)$.

*Proof.* First, note that the algorithm terminates finitely, since

$$0 \leqslant r_n < r_{n-1} < \cdots < r_1 < b,$$

a strictly decreasing sequence of nonnegative integers bounded below by 0 cannot continue indefinitely. Second, its correctness follows directly from Lemma 6.6. ■

**Example 6.8.**

$$\underbrace{49}_{a} = 2 \cdot \underbrace{21}_{b} + \underbrace{7}_{r_1}, \qquad 21 = 3 \cdot 7 + 0 \quad \Longrightarrow \quad \gcd(49, 21) = 7.$$

**Theorem 6.9** (Bézout)**.** *For $a, b \in \mathbb{N}$, there exist $x, y \in \mathbb{Z}$ such that*

$$\gcd(a, b) = ax + by.$$

*Proof.* Let $r_i$ denote the remainders in the Euclid algorithm for $\gcd(a, b)$. We prove by induction a more general statement: for any $r_i$ there exist $x_i, y_i \in \mathbb{Z}$ such that

$$r_i = a \cdot x_i + b \cdot y_i.$$

1. Basis. Indeed,

$$r_1 = a - q_1 \cdot b = a \cdot x_1 + b \cdot y_1. \qquad (x_1 = 1,\ y_1 = -q_1).$$

2. Suppose that $r_i = a \cdot x_i + b \cdot y_i$ for all $i$.

3. We must prove that $r_{i+1} = a \cdot x_{i+1} + b \cdot y_{i+1}$. From Euclid's algorithm,

$$\begin{aligned} r_{i+1} &= r_{i-1} - q_{i+1} \cdot r_i = a \cdot x_{i-1} + b \cdot y_{i-1} - q_{i+1} \cdot (a \cdot x_i + b \cdot y_i) \\ &= a \cdot \left(x_{i-1} - q_{i+1} \cdot x_i\right) + b \cdot \left(y_{i-1} - q_{i+1} \cdot y_i\right) \\ &= a \cdot x_{i+1} + b \cdot y_{i+1}. \end{aligned}$$

The proof is complete. ∎

**Exercise 6.10.** Where is a mistake in the above proof? Fix it.

Just to entertain ourselves, here is another proof.

*Alternative proof.* Let $S = \{ax + by : x, y \in \mathbb{Z}\}$, and let $m$ be its smallest positive element. Such an element exists because $a \in S$.

Let $d = \gcd(a, b)$. Since $d$ divides both $a$ and $b$, it divides every element of $S$. Thus $d \mid m$, so $d \leqslant m$. For any $n \in S$, write

$$n = qm + r, \qquad 0 \leqslant r < m.$$

Since $r = n - qm \in S$, the minimality of $m$ forces $r = 0$. Hence $m$ divides every element of $S$, including $a$ and $b$. Therefore $m \leqslant d$.

It follows that $m = d$. Since $m \in S$, we have $\gcd(a, b) = ax + by$ for some $x, y \in \mathbb{Z}$. ∎

**Corollary 6.11.** *Let $a, b \in \mathbb{N}$. Any common factor of $a$ and $b$ also divides* $\gcd(a, b)$.

*Proof.* There exist $x, y \in \mathbb{Z}$ such that $\gcd(a, b) = ax + by$. Let $d$ be such that $d \mid a$ and $d \mid b$. Then

$$d \mid (ax + by) \quad \Longrightarrow \quad d \mid \gcd(a, b).$$

∎

## 3 Coprime natural numbers

**Definition 6.12.** Natural numbers $a, b$ are called *coprime* or *relatively prime* if $\gcd(a, b) = 1$.

**Example 6.13.** 7 and 17 are coprime, as well as 8 and 15.

**Corollary 6.14.** *The natural numbers $a, b$ are coprime if and only if there exist $x, y \in \mathbb{Z}$ such that*

$$ax + by = 1.$$

*Proof.* *($\Rightarrow$)* Suppose $a, b$ are coprime, i.e. $\gcd(a, b) = 1$. By the Bézout theorem, there exist $x, y \in \mathbb{Z}$ such that $\gcd(a, b) = ax + by$. This yields $ax + by = 1$.

*($\Leftarrow$)* Suppose there exist $x, y \in \mathbb{Z}$ with $ax + by = 1$. Let $d = \gcd(a, b)$. Then $d \mid a$ and $d \mid b$, so $d \mid (ax + by)$. Hence $d \mid 1$, which forces $d = 1$. Thus $a, b$ are coprime. ∎

**Theorem 6.15.** *Let $a, b, c \in \mathbb{Z}$.*

*1) If $\gcd(a, c) = 1$ and $a \mid bc$, then $a \mid b$.*

*2) If $p$ is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.*

*Proof.* 1) There exist $x, y \in \mathbb{Z}$ such that $1 = ax + cy$. Then

$$b = abx + bcy \quad \Longrightarrow \quad a \mid b.$$

2) Let $p \nmid a$. Then $\gcd(a, p) = 1$. So there exist $x, y \in \mathbb{Z}$ such that $1 = ax + py$. Then

$$b = abx + bpy \quad \Longrightarrow \quad p \mid b.$$

Therefore, either $p \nmid a$ and $p \mid b$ or $p \mid a$. ∎

## 4 Prime factorization

**Theorem 6.16** (Fundamental Theorem of Arithmetic)**.** *Let $n \in \mathbb{N}$ with $n \geqslant 2$. Then*

*(i) $n$ can be written as a product of primes.*

*(ii) This factorization is unique (up to the order of the factors).*

*Proof.* *(i)* We use strong induction on $n$. The case $n = 2$ is clear, since 2 is prime. Suppose every integer from 2 to $n$ can be written as a product of primes. If $n + 1$ is prime, we are done. Otherwise, $n + 1 = ab$ with $2 \leqslant a, b \leqslant n$. By the induction hypothesis, both $a$ and $b$ are products of primes, and so is $n + 1$.

*(ii)* Suppose

$$n = p_1 \cdots p_k = q_1 \cdots q_\ell$$

are two prime factorizations. By Theorem 6.15, $p_1$ divides some $q_j$. Since both are prime, $p_1 = q_j$. Cancel these equal factors and repeat. No factors can remain on only one side, since a nonempty product of primes is greater than 1. Thus the two factorizations have the same primes, up to order. ∎

**Definition 6.17.** A *canonical representation* of $n \in \mathbb{N}$ is

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}, \qquad \text{where } p_1 < p_2 < \cdots < p_k \text{ and } \alpha_1, \ldots, \alpha_k \in \mathbb{N}.$$

**Example 6.18.** $540 = 2^2 \cdot 3^3 \cdot 5^1$.

**Exercise 6.19.** Let $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$ and $m = p_1^{\beta_1} \cdots p_k^{\beta_k}$. What is $\gcd(m, n)$?

**Theorem 6.20.** *A positive integer is a perfect square if and only if every exponent in its prime factorization is even.*

*Proof.* If $n = m^2$ and $m = \prod_i p_i^{\beta_i}$, then $n = \prod_i p_i^{2\beta_i}$, so all exponents are even. Conversely, if every exponent of $n$ is even, write

$$n = \prod_i p_i^{2\beta_i} = \left( \prod_i p_i^{\beta_i} \right)^2 .$$

■

**Example 6.21.** The number $52800 = 2^6 \cdot 3 \cdot 5^2 \cdot 11$ is not a perfect square, because the prime exponents are not all even.

**Theorem 6.22.** *Let $n \in \mathbb{N}$. Then $\sqrt{n} \in \mathbb{Q}$ if and only if $n$ is a perfect square.*

*Proof.* $(\Rightarrow)$ *If only* direction.

Suppose $\sqrt{n} \in \mathbb{Q}$. Then we can represent $n$ as $\sqrt{n} = \frac{a}{b}$ with $a, b \in \mathbb{N}$ and $\gcd(a, b) = 1$. Then

$$nb^2 = a^2.$$

All exponents of $a^2$ and $b^2$ in their canonical representations are even. If $n$ had in its canonical representation at least one factor with an odd exponent, then we would have a contradiction, since

$$\text{odd} + \text{even} \neq \text{even}.$$

$(\Leftarrow)$ *If* direction.

The reverse is immediate: if $n = m^2$ for some $m \in \mathbb{N}$, then $\sqrt{n} = m \in \mathbb{N} \subset \mathbb{Q}$. ■

**Exercise 6.23.** Prove that $6n + 5$ and $4n + 3$ are coprime for every $n \in \mathbb{N}$.

**Exercise 6.24.** Find all integer solutions of $42x + 30y = 18$

**Exercise 6.25.** Let $a, b \in \mathbb{N}$ with $\gcd(a, b) = 1$. Prove that $\gcd(a + b, a^2 + b^2) \in \{1, 2\}$.

**Exercise 6.26.** Let $n = 2^4 3^3 5^2$. How many positive divisors $d$ of $n$ satisfy $\gcd(d, n/d) = 1$?

# 7 Congruences

Yesterday, in our Julia tutorial, we had to solve the following problem: "Does there exist a power of 13 whose decimal representation ends in 33?" This was the approach we considered:

```
n = 100; a = 13;
for i in 1:n
    if rem(a, 100) == 33
        println("Yes! Here is an example: 13 to the power $i ends in 33")
        break
    end
    a = 13 * a
end
```

This code is correct, but it is not ideal. The reason is that the multiplication `a = 13 * a` inside the loop may give an enormous number[1] that Julia's standard integer type cannot represent, and the value will *overflow*. In this lecture, we will learn how to fix the code above.

## 1 Definition and elementary properties

Consider addition and multiplication of odd and even numbers.

**Addition**

| $+$ | $o$ | $e$ |
|---|---|---|
| $o$ | $e$ | $o$ |
| $e$ | $o$ | $e$ |

**Multiplication**

| $\times$ | $o$ | $e$ |
|---|---|---|
| $o$ | $o$ | $e$ |
| $e$ | $e$ | $e$ |

Using these tables, it is easy to determine the parity of the sum $213123891283902123 + 2^{100}$ without computing the actual sum. Here the actual numbers do not matter; only their parity does.

**Definition 7.1.** Let $a, b, m \in \mathbb{Z}$, $m > 1$. We say that $a$ is *congruent* to $b$ *modulo* $m$ if

$$m \mid a - b.$$

We write $a \equiv b \pmod{m}$, or sometimes $a \equiv b\ (m)$.

**Example 7.2.** Let $m = 5$ and $a = 7$. Verify that $12 \equiv a \pmod 5$, $17 \equiv a \pmod 5$, and $132 \equiv a \pmod 5$.

**Lemma 7.3.** *$a \equiv b \pmod m$ if and only if $a$ and $b$ have the same remainder upon division by $m$.*

*Proof.* If $a \equiv b \pmod m$, then $a = q_1 m + r_1$ and $b = q_2 m + r_2$ with $0 \leqslant r_1 < m$, $0 \leqslant r_2 < m$. Thus,

$$a - b = (q_1 - q_2)m + (r_1 - r_2).$$

Since $m \mid a - b$, it must hold that $m \mid r_1 - r_2$, which forces $r_1 - r_2 = 0$.

The reverse direction is immediate. ∎

[1] This particular code will terminate before that happens, but in principle it is still bad practice.

**Theorem 7.4.** *Let $a, b, m \in \mathbb{Z}$, $m > 1$. Then*

*(i)* $a \equiv a \pmod{m}$.

*(ii)* $a \equiv b \pmod{m} \iff b \equiv a \pmod{m}$.

*(iii)* $a \equiv b \pmod{m}$ *and* $b \equiv c \pmod{m} \implies a \equiv c \pmod{m}$.

**Example 7.5.** It is currently 10:00 in the morning. What time will it be in 27 hours? We would not answer 37:00. Instead, we perform addition modulo 24:

$$10 + 27 \equiv 13 \pmod{24}$$

and answer 13:00. If we used the 12-hour format instead, we would perform addition modulo 12:

$$10 + 27 \equiv 1 \pmod{12},$$

which corresponds to 1 p.m.

**Exercise 7.6.** It is now 17:00. Can you see the Sun in 54 hours?

**Theorem 7.7.** *Let $a, b, c, d \in \mathbb{Z}$, $m > 1$. Then*

*1)* $a \equiv c \pmod{m}$ *and* $b \equiv d \pmod{m} \implies a + b \equiv c + d \pmod{m}$.

*2)* $a \equiv c \pmod{m}$ *and* $b \equiv d \pmod{m} \implies ab \equiv cd \pmod{m}$.

*Proof.* **1)** $m \mid a - c$ and $m \mid b - d \implies m \mid (a + b) - (c + d) \implies a + b \equiv c + d \pmod{m}$.

**2)** There exist $x, y \in \mathbb{Z}$ such that $a - c = x \cdot m$ and $b - d = y \cdot m$. Then

$$ab = (mx + c)(my + d) = m^2xy + mxd + mcy + cd.$$

Since $m \mid \left(m^2xy + mxd + mcy\right)$, it follows that $ab \equiv cd \pmod{m}$. ∎

**Example 7.8.** Let $a = 4$, $b = 5$, $c = 10$, $d = 11$, $m = 3$.

$$a + b = 9, \qquad c + d = 21, \qquad 9 \equiv 21 \pmod{3}.$$

$$ab = 20, \qquad cd = 110, \qquad 20 \equiv 110 \pmod{3}.$$

**Example 7.9.** Does there exist $x \in \mathbb{Z}$ such that $x^2 - 51x - 123 = 0$?

Of course, this is a quadratic equation, and one can simply find its roots by computing the discriminant. Here we will illustrate another idea.

*Solution.* Consider this equation modulo 2. If $x \equiv 0 \pmod{2}$, then $x^2 - 51x - 123 \equiv 1 \pmod{2}$, and this cannot be 0. If $x \equiv 1 \pmod{2}$, then $x^2 - 51x - 123 \equiv 1 \pmod{2}$, and this again cannot be 0. Hence, there is no such integer $x$. ◇

**Corollary 7.10.** *Let $n \in \mathbb{N}$. Then* $a \equiv b \pmod{m} \implies a^n \equiv b^n \pmod{m}$.

**Exercise 7.11.** Does the converse hold in general?

## 2 Applications

**Proposition 7.12** (Divisibility test for 3)**.** *A number $n \in \mathbb{N}$ is divisible by* 3 *if and only if the sum of its digits is divisible by* 3*.*

*Proof.* Let $n = (a_k a_{k-1} \dots a_1 a_0)_{10}$. Note that $10^i \equiv 1 \pmod 3$ for any $i$. Hence, we have

$$n = \sum_{i=0}^{k} a_i \cdot 10^i \equiv \sum_{i=0}^{k} a_i \pmod 3.$$

In other words, $n$ and the sum of its digits have the same remainder upon division by 3. ∎

**Exercise 7.13.** Do you see why the same test also holds for 9? Can you derive a similar test for 11?

**Example 7.14** (Problem)**.** What is the remainder when $6^{30}$ is divided

1) by 5?
2) by 7?
3) by 13?

*Solution.* **1)** We have

$$6 \equiv 1 \pmod 5 \quad \Longrightarrow \quad 6^{30} \equiv 1^{30} \equiv 1 \pmod 5.$$

**2)** We have

$$6 \equiv -1 \pmod 7 \quad \Longrightarrow \quad 6^{30} \equiv (-1)^{30} \equiv 1 \pmod 7.$$

**3)** Let's compute successive powers of 6 modulo 13:

$$\begin{aligned} 6^1 &\equiv 6, \\ 6^2 &\equiv -3, \\ 6^3 &\equiv -5, \\ 6^4 &\equiv -4, \\ 6^5 &\equiv -24 \equiv 2, \\ 6^6 &\equiv -1, \\ 6^{12} &\equiv 1, \\ 6^{13} &\equiv 6 \pmod{13}. \end{aligned}$$

Hence

$$6^{30} = 6^{24} \cdot 6^6 = (6^{12})^2 \cdot 6^6 \equiv 1^2 \cdot (-1) \equiv -1 \pmod{13}.$$

◇

**Example 7.15.** Solve $x^2 - 3y = 2$ in $\mathbb{Z}$.

*Solution.* Consider the equation modulo 3. First, note that $x^2 - 3y \equiv x^2 \pmod 3$. The possible remainders are

| $x$ | $x^2$ |
|---|---|
| 0 | 0 |
| 1 | 1 |
| 2 | 1 |

The left-hand side has remainder 0 or 1, whereas the right-hand side has remainder 2. Hence, there is no such integer $x$. ◇

## 3 Residue classes and $\mathbb{Z}_m$

**Definition 7.16.** Let $a \in \mathbb{Z}$ and $m \in \mathbb{Z}$ with $m > 1$. The set

$$\overline{a} = \{n \in \mathbb{Z} \,:\, a \equiv n \pmod{m}\}$$

is called the *congruence class* (or *residue class*) of $a$ modulo $m$.

**Example 7.17.** Let $m = 3$. Then $\overline{1} = \{\dots, -5, -2, 1, 4, 7, \dots\}$ is the set of all integers that have remainder 1 upon division by 3. Similarly,

$$\overline{2} = \{\dots, -4, -1, 2, 5, 8, \dots\}, \qquad \overline{0} = \{\dots, -6, -3, 0, 3, 6, 9, \dots\}, \qquad \overline{4} = \overline{1}.$$

For general $m$, there are exactly $m$ distinct classes

$$\overline{0},\ \overline{1},\ \overline{2},\ \dots,\ \overline{m-1}.$$

$\mathbb{Z}_m$ is the set of all congruence classes modulo $m$.

**Arithmetic in $\mathbb{Z}_m$.** We can define our own arithmetic rules in $\mathbb{Z}_m$:

$$\overline{a} + \overline{b} = \overline{a+b}, \qquad \overline{a} - \overline{b} = \overline{a-b}, \qquad \overline{a} \cdot \overline{b} = \overline{ab}.$$

**Example 7.18.** Let $m = 5$.

$$\overline{2} + \overline{3} = \overline{0}, \qquad \overline{2} + \overline{4} = \overline{1}, \qquad \overline{3} \cdot \overline{4} = \overline{2}.$$

We have to ensure that our arithmetic rules are well defined. In the example above, we have to check, for instance, that $\overline{2} + \overline{3}$ and $\overline{2} + \overline{8}$ lead to the same result. In general, if $a \equiv a' \pmod{m}$ and $b \equiv b' \pmod{m}$, then the compatibility properties proved above give

$$a + b \equiv a' + b', \qquad ab \equiv a'b' \pmod{m}.$$

Also, $m \mid (a - a')$ and $m \mid (b - b')$ imply $m \mid \big((a-b) - (a'-b')\big)$. Thus, all three operations are independent of the chosen representatives.

Look at what we have done: both $\overline{2}$ and $\overline{3}$ are sets. Ordinary integer addition and multiplication do not directly apply to sets. For these particular sets, however, we have defined arithmetic operations using representatives and shown that the results do not depend on our choices. This illustrates a powerful idea in mathematics: identifying objects that are equivalent for a particular purpose. Modulo 5, the numbers 1, 6, and 181 are equivalent. We put them in the same class and can calculate using any one of its representatives.

**Visa/Mastercard checksum.** If you have ever used your bank card online, you may have noticed that if you enter your card number incorrectly, the website can flag the error immediately — even before contacting your bank. This is possible because of a checksum calculated using the Luhn algorithm.

LUHN'S ALGORITHM

1. Count the rightmost digit as position 1.
2. Double the digits in positions $2, 4, 6, \ldots$.
3. If a doubled value has two digits, replace it by the sum of its digits.
4. Add the resulting values and the unchanged digits.
5. Accept the checksum if the total is divisible by ten.

For example, let's check the illustrative digit string 7623457:

| Original digit | 7 | 6 | 2 | 3 | 4 | 5 | 7 |
|---|---|---|---|---|---|---|---|
| Position from right | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| Action | Keep | Double | Keep | Double | Keep | Double | Keep |
| Result | 7 | 3 | 2 | 6 | 4 | 1 | 7 |

The resulting sum satisfies

$$7 + 3 + 2 + 6 + 4 + 1 + 7 = 30 \equiv 0 \pmod{10}.$$

Hence, this digit string passes the Luhn checksum test.

**Exercise 7.19.** Why did we not simply check the sum of all the original digits modulo 10?

**Exercise 7.20.** Verify the checksum of your own bank card.

**Exercise 7.21.** Fix the Julia code at the beginning of this lecture.

**Exercise 7.22.** When are the congruences $a \equiv b \pmod{m}$ and $ac \equiv bc \pmod{m}$ equivalent?

**Exercise 7.23.** There are 100 stones. Two players take turns removing between 1 and 5 stones. The player who takes the last stone loses. Determine a winning strategy for the first player.

**Exercise 7.24.** Find the remainders when $2^{2026}$ is divided by $3, 5, 7, \ldots, 13$.

**Exercise 7.25.** Let $a, b \in \mathbb{Z}$. Prove that $3 \mid a^2 + b^2$ implies $9 \mid a^2 + b^2$.

**Exercise 7.26.** Prove that

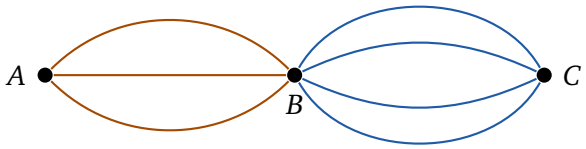
$$7 \mid 2222^{5555} + 5555^{2222}.$$

# 8 Counting

## 1 Sum and product rules

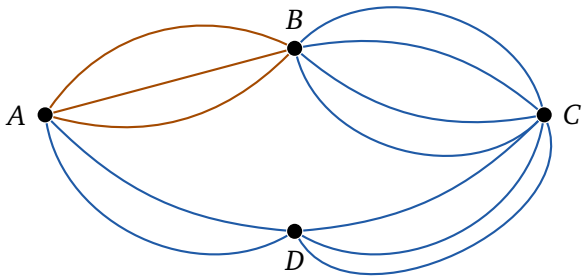
Consider the following problem: There are three roads from $A$ to $B$ and four roads from $B$ to $C$. In how many ways can one travel from $A$ to $C$, as shown in the picture?

**Answer:** $3 \times 4 = 12$.



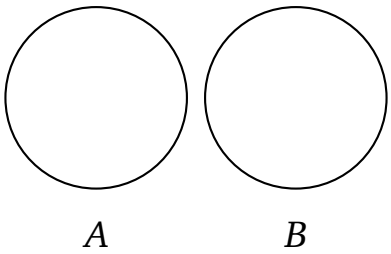

Now add another city, $D$, to the previous setting. There are 2 roads from $A$ to $D$ and 3 roads from $D$ to $C$. The question remains the same: In how many ways can one travel from $A$ to $C$, as shown in the picture?

**Answer:** $3 \times 4 + 2 \times 3 = 18$.




Let's generalize these problems.

**Sum rule.** If there are $m$ ways to choose an element $a$ and $n$ ways to choose an element $b$ (independent of $a$), then there are $m + n$ ways to choose "$a$ or $b$".




$$|A \cup B| = |A| + |B|.$$

**Product rule.** If there are $m$ ways to choose an element $a$ and $n$ ways to choose an element $b$, then there are $mn$ ways to choose "$a$ and $b$".

$$|A \times B| = |A| \cdot |B|.$$

**Example 8.1.** How many 3-digit numbers are there?

$$\underset{9}{\bullet} \quad \underset{10}{\bullet} \quad \underset{10}{\bullet} \qquad 9 \cdot 10 \cdot 10 = 900.$$

**Example 8.2.** How many 6-digit numbers are there that are divisible by 5?

$$9 \cdot 10^4 \cdot 2 = 18 \cdot 10^4.$$

**Example 8.3.** How many 6-digit numbers are there that contain at least one even digit?

*Solution.* There are $9 \cdot 10^5$ 6-digit numbers, and there are $5^6$ numbers that contain only odd digits. Hence, there are $9 \cdot 10^5 - 5^6$ 6-digit numbers that contain at least one even digit. ◇

**Theorem 8.4.** *A set of $n$ elements can be arranged in $n! = 1 \cdot 2 \cdots n$ different orders.*

Alternatively, there are $n!$ permutations of a list of $n$ distinct elements. Can you prove this? Since $0! = 1$ by convention, the theorem is also valid for the empty set.

## 2 Ordered and unordered selections

There are 200 students in the class. Consider two problems:

1. You have to choose three students who will go to the meeting with the Rector.
2. You have to choose three students who will serve as president, vice-president, and secretary at the student conference.

In how many ways can you do this in each case? What is the essential difference between the two cases?

**Theorem 8.5.** *Let $n, m$ be integers with $0 \leqslant m \leqslant n$, and let $S$ be a set of $n$ elements. The number of ordered selections of $m$ distinct elements from $S$ is*

$$P(n,m) = n(n-1)\cdots(n-m+1) = \frac{n!}{(n-m)!}.$$

**Theorem 8.6.** *Let $n, m$ be integers with $0 \leqslant m \leqslant n$, and let $S$ be a set of $n$ elements. The number of unordered selections of $m$ distinct elements from $S$ is*

$$\binom{n}{m} = \frac{P(n,m)}{m!} = \frac{n!}{m!(n-m)!}.$$

The notation $\binom{n}{m}$ is read as "$n$ choose $m$."
**Alternative notation.** Sometimes $C_n^m$ or $C(n,m)$ is used instead of $\binom{n}{m}$.
For $n \geqslant 2$, let's compute $\binom{n}{m}$ for several values of $m$:

$$\binom{n}{0} = 1, \quad \binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2}, \quad \ldots \quad \binom{n}{n-2} = \frac{n(n-1)}{2}, \quad \binom{n}{n-1} = n, \quad \binom{n}{n} = 1.$$

What you see is not a coincidence.

**Theorem 8.7.** *For $0 \leqslant m \leqslant n$,*

$$\binom{n}{m} = \binom{n}{n-m}.$$

*Proof.* An algebraic proof is immediate. Can you find a combinatorial proof? ■

**Example 8.8.** At a party with $n$ people, everyone shook hands with every other person exactly once. How many handshakes were there?

*Solution.* $\binom{n}{2} = \frac{n(n-1)}{2}$. ◇

**Example 8.9.** There are 2 mathematics professors and 8 computer science professors. We need to form a committee of 6 people, at least 1 of whom must be a mathematics professor. In how many ways can we form the committee?

*Solution.* We have two cases: either there is 1 mathematics professor on the committee or there are 2. These two cases are mutually exclusive. Hence, the answer is

$$\binom{2}{1} \cdot \binom{8}{5} + \binom{2}{2} \cdot \binom{8}{4}.$$

Alternatively, we can argue as follows: In total, there are $\binom{10}{6}$ six-member committees, and among them there are $\binom{8}{6}$ that do not include a mathematics professor. Therefore, there must be

$$\binom{10}{6} - \binom{8}{6}$$

committees that satisfy the condition.

We obtained two seemingly very different answers. But if you compute the actual numbers, they will be the same. ◇

**Example 8.10.** Alice has 6 different chocolate candies, and Bob has 8 different gummy bears. In how many ways can they each choose 5 of their items to exchange?

*Solution.*

$$\binom{6}{5} \cdot \binom{8}{5}.$$

◇

**Example 8.11.** There are 12 people, including Alice and Bob. In how many ways can one choose a committee of 6 people under each of the following conditions?

(i) There are no restrictions.

(ii) Both Alice and Bob are members.

(iii) Exactly one of Alice and Bob is a member.

*Solution.* The answers are

$$\binom{12}{6}, \qquad \binom{10}{4}, \qquad 2\binom{10}{5}$$

respectively. Explain why. ◇

## 3 Binomial coefficients

You may remember these formulas:

$$(x+y)^2 = (x+y)(x+y) = x^2 + 2xy + y^2$$
$$(x+y)^3 = (x+y)(x+y)(x+y) = x^3 + 3x^2y + 3xy^2 + y^3$$

Now we would like to generalize them to an arbitrary power $n \in \mathbb{N}$.

**Theorem 8.12** (Binomial theorem)**.** *For $n \in \mathbb{N}$,*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

*Proof.* It should be clear that

$$(x+y)^n = (x+y)\ldots(x+y) = \text{sum of terms of the form } x^{n-k}y^k \text{ for different } k,$$

so the question is how many copies of each term $x^{n-k}y^k$ occur, or, equivalently, what the coefficient of each such term is. In fact, the coefficient of $x^{n-k}y^k$ must be $\binom{n}{k}$, because there are $n$ factors in total and we must choose $k$ of them from which to take the term $y$. ■

This is why we refer to the numbers $\binom{n}{k}$ as binomial coefficients.

**Example 8.13.** What is the coefficient of $x^5y^8$

a) in $(x+y)^{12}$?

b) in $(x+y)^{13}$?

*Solution.* a) 0. b) $\binom{13}{5}$ or $\binom{13}{8}$. ◇

PASCAL'S TRIANGLE

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $n=0$ | | | | | | 1 | | | | | | |
| 1 | | | | | 1 | | 1 | | | | | |
| 2 | | | | 1 | | 2 | | 1 | | | | |
| 3 | | | 1 | | (3) | | (3) | | 1 | | | |
| 4 | | 1 | | 4 | | (6) | | 4 | | 1 | | |
| 5 | 1 | | 5 | | 10 | | 10 | | 5 | | 1 | |

Every element in the triangle equals the sum of the two neighbors above it: $6 = 3 + 3$.

**Theorem 8.14** (Pascal's identity)**.** *For $1 \leqslant m \leqslant n$,*

$$\binom{n}{m} = \binom{n-1}{m-1} + \binom{n-1}{m}.$$

*Proof (algebraic).*

$$\begin{aligned}\text{RHS} &= \frac{(n-1)!}{(m-1)!(n-m)!} + \frac{(n-1)!}{m!(n-m-1)!} \\ &= \frac{(n-1)!}{(m-1)!(n-m-1)!}\left(\frac{1}{n-m} + \frac{1}{m}\right) \\ &= \frac{(n-1)!}{(m-1)!(n-m-1)!} \cdot \frac{n}{(n-m)\cdot m} \\ &= \frac{n!}{m!(n-m)!} = \binom{n}{m}\end{aligned}$$

■

*Proof (combinatorial).* Let $S = \{a_1, \ldots, a_n\}$. Then $\binom{n}{m}$ is the number of subsets of $S$ of cardinality $m$. Consider an element $a_1$. There are

- $\binom{n-1}{m-1}$ subsets of cardinality $m$ that contain $a_1$;
- $\binom{n-1}{m}$ subsets of cardinality $m$ that do not contain $a_1$.

Any subset either contains $a_1$ or does not. Hence,

$$\binom{n-1}{m-1} + \binom{n-1}{m} = \binom{n}{m}.$$

■

**Corollary 8.15.**

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

*Proof.* Set $x = y = 1$ in the binomial theorem. Combinatorially, both sides count all subsets of an $n$-element set. ■

# 4 Sticks and stones

Consider the problem: How many solutions does $x + y + z = 10$ have in $\mathbb{N} \cup \{0\}$?

Let us represent the number 10 as ten stones, and use two sticks as separators to divide them into three parts for $x$, $y$, and $z$:

$$\underbrace{\bullet\bullet\bullet\bullet}_{x=4} \mid \underbrace{\bullet\bullet\bullet\bullet\bullet\bullet}_{y=5} \mid \underbrace{\bullet}_{z=1}$$

For example, the arrangement $\bullet \,\|\, \bullet\bullet\bullet\bullet\bullet\bullet\bullet\bullet\bullet$ corresponds to:

$$x = 1, \quad y = 0, \quad z = 9.$$

Clearly, for every $(x, y, z)$ there is an arrangement of ten stones and two sticks, and vice versa. Hence, we must choose the positions of two sticks among 12 positions in total (ten stones and

two sticks). Thus, the answer is

$$\binom{12}{2}.$$

**Exercise 8.16.** How many solutions does $x + y + z = 10$ have in $\mathbb{N}$?

**Exercise 8.17.** How many nondecreasing sequences of length six can be made using only 1, 2, 3, 4? For example, 1, 1, 2, 2, 2, 4 is allowed.

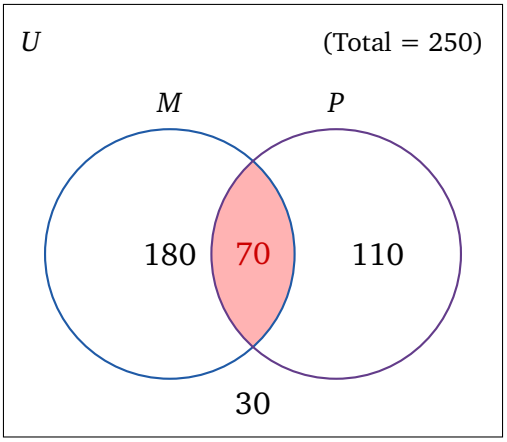
**Exercise 8.18.** How many five-digit numbers have digits that never decrease from left to right? For example, 11247 satisfies the condition.

## 5 Inclusion–exclusion principle

Consider the problem:

> There are 250 students in total. Of these, 180 chose the program because of mathematics, 110 because of programming, and 70 because of both mathematics and programming. How many students chose the program for neither reason?

*Solution.* We need to find $|(M \cup P)^c|$.




We have

$$|M \cup P| = |M| + |P| - |M \cap P| = 180 + 110 - 70 = 220.$$

Hence, there are $250 - 220 = 30$ students who chose the program for neither reason. ◇

**Theorem 8.19** (Inclusion-exclusion for 2 sets)**.** *For finite sets $A$ and $B$,*

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

**Theorem 8.20** (Inclusion-exclusion for 3 sets)**.** *For finite sets $A$, $B$, and $C$,*

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

*Proof.* Every $x \in A \cup B \cup C$ belongs to exactly one, exactly two, or all three of the sets:

- If $x$ belongs to exactly one set, the RHS counts it with +1;
- if $x$ belongs to exactly two sets, the RHS counts it with $+1 + 1 - 1 = 1$;
- if $x$ belongs to all three sets, the RHS counts it with $+1 + 1 + 1 - 1 - 1 - 1 + 1 = 1$.

The RHS counts no elements outside $A \cup B \cup C$, and it counts every element in the union exactly once. This proves the formula. ■

This formula extends to $n$ sets by a similar argument, but the general expression is cumbersome, so we omit it here.

**Exercise 8.21.** How many natural numbers $n$ with $n \leqslant 1000$ are divisible by 7 or 11?

*Solution.* Let the universal set be $U = \{1, 2, \ldots, 1000\}$. Define the sets of multiples as follows:

$$A = \{n \leqslant 1000 : 7 \mid n\}, \quad B = \{n \leqslant 1000 : 11 \mid n\}$$

We want to find the cardinality of the union $|A \cup B|$.

We use the notation $\lfloor x \rfloor$ for the largest integer that does not exceed $x$ (the integer part).

i) Count multiples of 7:

$$|A| = \left\lfloor \frac{1000}{7} \right\rfloor = 142 \qquad \text{(because } 7 \cdot 1,\ 7 \cdot 2,\ \ldots,\ 7 \cdot 142\text{)}$$

ii) Count multiples of 11:

$$|B| = \left\lfloor \frac{1000}{11} \right\rfloor = 90 \qquad \text{(because } 11 \cdot 1,\ 11 \cdot 2,\ \ldots,\ 11 \cdot 90\text{)}$$

iii) Count multiples of both (intersection): Since $\gcd(7, 11) = 1$, a number is divisible by both 7 and 11 if and only if it is divisible by 77. Then $|A \cap B| = \{n \leqslant 1000 : 77 \mid n\}$ and similarly

$$|A \cap B| = \left\lfloor \frac{1000}{77} \right\rfloor = 12 \qquad \text{(because } 77 \cdot 1,\ 77 \cdot 2,\ \ldots,\ 77 \cdot 12\text{)}$$

Now, applying the principle of inclusion–exclusion, we get

$$|A \cup B| = |A| + |B| - |A \cap B| = 142 + 90 - 12 = 220.$$

◇

# 9 Relations

## 1 Relations

**Definition 9.1.** Let $A, B$ be sets. A *relation from $A$ to $B$* is a subset

$$R \subset A \times B.$$

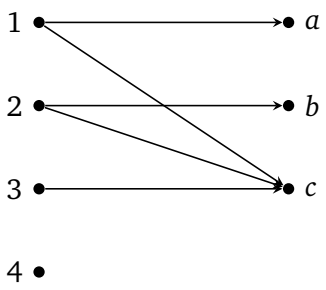
Notation: $(a, b) \in R$ or $aRb$ and $(a, b) \notin R$ or $a \not R b$.
If $A = B$, then $R$ is called *a relation on $A$*.

**Example 9.2.** Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c\}$. Define the relation $R \subset A \times B$ as:

$$R = \{(1, a), (1, c), (2, b), (2, c), (3, c)\}$$

We can illustrate it using arrows or a table.

**Arrow diagram**




**Table**

| | $a$ | $b$ | $c$ |
|---|---|---|---|
| 1 | + | | + |
| 2 | | + | + |
| 3 | | | + |
| 4 | | | |

**Example 9.3.** Define the relation $R$ on $A = \{1, 2, 3, 4\}$ as $a \, R \, b \iff a < b$.
In explicit set notation, this gives:

$$R = \{(1, 2),\ (1, 3),\ (1, 4),\ (2, 3),\ (2, 4),\ (3, 4)\}$$

**Example 9.4.** Define the relation $R$ on $A = \{1, 2, 3, 4\}$ as $a \, R \, b \iff a \mid b$.
In explicit set notation, this gives:

$$R = \{(1, 1),\ (1, 2),\ (1, 3),\ (1, 4),\ (2, 2),\ (2, 4),\ (3, 3),\ (4, 4)\}$$

## 2 Properties

**Definition 9.5.** Let $R$ be a relation on a set $S$. The relation $R$ is:

- *reflexive*: if $aRa$ for every $a \in S$;
- *symmetric*: if $aRb \implies bRa$;
- *antisymmetric* if $aRb$ and $bRa \implies a = b$;
- *transitive* if $aRb$ and $bRc \implies aRc$.

**Exercise 9.6.** Which of the following relations on $S = \{1, 2, 3, 4\}$ are reflexive, symmetric, antisymmetric, or transitive?

$$R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\},$$
$$R_2 = \{(1,1), (1,2), (2,1)\},$$
$$R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\},$$
$$R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\},$$
$$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\},$$
$$R_6 = \{(3,4)\}.$$

**Exercise 9.7.** Can a relation be symmetric and antisymmetric at the same time?

**Example 9.8.** Let $S$ be a set with $|S| = n$. How many relations are there on $S$?

*Solution.* A relation on $S$ is defined as any subset of the Cartesian product $S \times S$. Since $|S \times S| = |S| \cdot |S| = n^2$, the total number of relations on $S$ is equal to the cardinality of the power set $\mathcal{P}(S \times S)$:

$$|\mathcal{P}(S \times S)| = 2^{n^2}$$

◇

**Exercise 9.9.** Let $S$ be a set with $|S| = n$. How many reflexive relations are there on $S$? How many symmetric relations are there on $S$?

**Example 9.10.** Consider the following relations $R$ on sets $S$ and their properties:

1. $S = \mathbb{Z}$ and $aRb \iff a - b$ is even: reflexive, symmetric, transitive;
2. $S = \mathbb{N}$ and $aRb \iff a \mid b$: reflexive, antisymmetric, transitive, but not symmetric;
3. $S = \mathbb{R}$ and $aRb \iff |a - b| \leqslant 1$: reflexive and symmetric, but not transitive.

## 3 Equivalence relations

**Definition 9.11.** An *equivalence relation* is a relation that is reflexive, symmetric, and transitive.

Notation: very often the equivalence is denoted by $\sim$, that is $aRb \iff a \sim b$.

Examples

1. **Absolute value equality on integers**

   Let $S = \mathbb{Z}$. Define the relation $R$ by $aRb \iff |a| = |b|$.

   - **Reflexive:** For all $a \in \mathbb{Z}$, $|a| = |a|$ holds, so $aRa$.
   - **Symmetric:** If $aRb$, then $|a| = |b|$, which implies $|b| = |a|$, so $bRa$.
   - **Transitive:** If $aRb$ and $bRc$, then $|a| = |b|$ and $|b| = |c|$. Thus, $|a| = |c|$, so $aRc$.

   **Conclusion:** Yes, $R$ is an equivalence relation.

2. **Integer difference on real numbers**

   Let $S = \mathbb{R}$. Define the relation $R$ by $aRb \iff a - b \in \mathbb{Z}$.

- **Reflexive:** For all $a \in \mathbb{R}$, $a - a = 0 \in \mathbb{Z}$, so $aRa$.
- **Symmetric:** If $aRb$, then $a - b = k$ for some $k \in \mathbb{Z}$. Then $b - a = -k \in \mathbb{Z}$, so $bRa$.
- **Transitive:** If $aRb$ and $bRc$, then $a - b \in \mathbb{Z}$ and $b - c \in \mathbb{Z}$. Summing these gives:

$$(a - b) + (b - c) = a - c \in \mathbb{Z}$$

  Therefore, $aRc$.

**Conclusion:** Yes, $R$ is an equivalence relation.

3. **Modular congruence on integers**

   Let $S = \mathbb{Z}$, $m \in \mathbb{N}$ with $m \geqslant 2$. Define $aRb \iff a \equiv b \pmod{m}$.

   - **Reflexive:** For all $a \in \mathbb{Z}$, $m \mid (a - a)$ since $a - a = 0 = 0 \cdot m$, so $aRa$.
   - **Symmetric:** If $a \equiv b \pmod{m}$, then $m \mid (a - b)$. Thus, $m \mid -(a - b) = (b - a)$, so $b \equiv a \pmod{m}$.
   - **Transitive:** If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $m \mid (a - b)$ and $m \mid (b - c)$. Hence, $m \mid \big((a - b) + (b - c)\big) = (a - c)$, so $a \equiv c \pmod{m}$.

   **Conclusion:** Yes, $R$ is an equivalence relation.

4. **Divisibility relation on natural numbers**

   Let $S = \mathbb{N}$. Define $aRb \iff a \mid b$.

   - **Reflexive:** For all $a \in \mathbb{N}$, $a \mid a$ since $a = 1 \cdot a$.
   - **Symmetric:** If $a \mid b$, it does not guarantee $b \mid a$. For example, $2 \mid 4$, but $4 \nmid 2$.
   - **Transitive:** doesn't matter.

   **Conclusion:** No, $R$ is not an equivalence relation.

5. **Bounded difference on real numbers**

   Let $S = \mathbb{R}$. Define $aRb \iff |a - b| < 1$.

   - **Reflexive:** For all $a \in \mathbb{R}$, $|a - a| = 0 < 1$, so $aRa$.
   - **Symmetric:** If $|a - b| < 1$, then $|b - a| = |a - b| < 1$, so $bRa$.
   - **Transitive:** Consider $a = 0$, $b = 1/2$, and $c = 1$.

$$|a - b| = \left|0 - \frac{1}{2}\right| = \frac{1}{2} < 1 \quad (aRb)$$

$$|b - c| = \left|\frac{1}{2} - 1\right| = \frac{1}{2} < 1 \quad (bRc)$$

     However, $|a - c| = |0 - 1| = 1 \not< 1$, so $a \not R c$.

   **Conclusion:** No, $R$ is not an equivalence relation.

6. **Sibling relation on people**

   Let $S$ = PEOPLE. Define $aRb \iff a$ is a sibling of $b$.

- **Reflexive:** A person is not considered their own sibling ($a \not R a$).
- **Symmetric:** Doesn't matter.
- **Transitive:** Doesn't matter.

**Conclusion:** No, $R$ is not an equivalence relation.

7. **Connectivity relation on cities**

   Let $S$ be the set of all cities. Define $aRb$ if there is a railway route from $a$ to $b$, allowing a route of length zero.

   - **Reflexive:** The route of length zero connects each city to itself.
   - **Symmetric:** A route from $a$ to $b$ can be traversed in reverse from $b$ to $a$.
   - **Transitive:** A route from $a$ to $b$ can be followed by a route from $b$ to $c$, giving a route from $a$ to $c$.

   **Conclusion:** Yes, $R$ is an equivalence relation.

8. **Triangles in the plane**

   Let $S$ be the set of all triangles in the plane. Define $aRb$ if and only if triangles $a$ and $b$ are similar (their corresponding angles are equal).

   - **Reflexive:** Every triangle is similar to itself.
   - **Symmetric:** If $aRb$, then $bRa$.
   - **Transitive:** If a triangle $a$ is similar to $b$, and $b$ is similar to $c$, then $a$ is similar to $c$.

**Definition 9.12.** If $R$ is an equivalence relation on $S$, the *equivalence class* of $a \in S$ is

$$[a] = \{s \in S \,:\, aRs\}.$$

**Example 9.13.** For congruence modulo $m$,

$$aRb \iff a \equiv b \pmod{m},$$

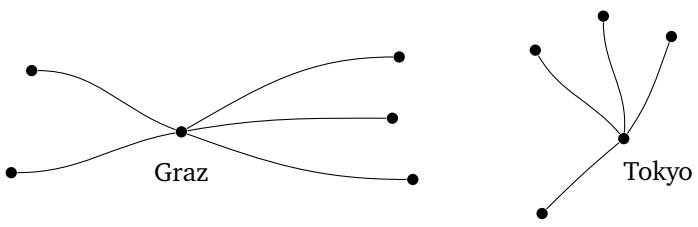
we have that

$$[1] = \overline{1} = \{n \colon n \equiv 1 \pmod{m}\}$$

and similarly for other classes. The equivalence classes are the residue classes

$$[0], [1], \ldots, [m-1].$$

**Example 9.14.** In the example with cities connected by railway, here are two classes:



**Theorem 9.15.** *Let $R$ be an equivalence relation on $S$, and let $a, b \in S$. Then the following are equivalent:*

*(i)* $aRb$

*(ii)* $[a] = [b]$

*(iii)* $[a] \cap [b] \neq \varnothing$

*Proof.* We will prove the implication chain (i) $\Longrightarrow$ (ii) $\Longrightarrow$ (iii) $\Longrightarrow$ (i).

**(i) $\Longrightarrow$ (ii):** Suppose $aRb$. We first show that $[a] \subset [b]$.

Let $s \in [a]$, which means $(a, s) \in R$ or equivalently $aRs$. Since $aRb$, by symmetry we have $bRa$. Now combining $bRa$ and $aRs$, transitivity gives $bRs$, which means $s \in [b]$. Hence, $[a] \subset [b]$.

By symmetry of roles, $[b] \subset [a]$. Therefore, $[a] = [b]$.

**(ii) $\Longrightarrow$ (iii):** Obvious, because $[a] \neq \varnothing$ (since $a \in [a]$ by reflexivity). Thus, $[a] \cap [b] = [a] \cap [a] = [a] \neq \varnothing$.

**(iii) $\Longrightarrow$ (i):** We have $[a] \cap [b] \neq \varnothing$, and we must show $aRb$.

Since $[a] \cap [b] \neq \varnothing$, there exists $s \in [a] \cap [b]$. This implies:

$$aRs \quad \text{and} \quad bRs$$

By symmetry, $bRs$ implies $sRb$. From $aRs$ and $sRb$, by transitivity, it follows that $aRb$. ∎

**Definition 9.16.** A *partition* of $S$ is a family of nonempty, pairwise disjoint subsets whose union is $S$.

**Theorem 9.17.** *Let $R$ be an equivalence relation on $S$. Then the equivalence classes of $R$ form a partition of $S$.*

*Proof.* On one hand, for every $a \in S$, reflexivity gives $a \in [a]$, so the classes are nonempty and cover $S$. On the other hand, if $[a] \neq [b]$, then by Theorem 9.15, $[a] \cap [b] = \varnothing$. Thus any two different classes are disjoint, so they form a partition of $S$. ∎

**Example 9.18.** Let $S = \mathbb{Z}$ and define $xRy \iff x \equiv y \pmod 4$. The equivalence classes are:

$$\begin{aligned}
[0] &= \bar{0} = \{\ldots, -8, -4, 0, 4, 8, \ldots\} \\
[1] &= \bar{1} = \{\ldots, -7, -3, 1, 5, 9, \ldots\} \\
[2] &= \bar{2} = \{\ldots, -6, -2, 2, 6, 10, \ldots\} \\
[3] &= \bar{3} = \{\ldots, -5, -1, 3, 7, 11, \ldots\}
\end{aligned}$$

The set of integers is partitioned as:

$$\mathbb{Z} = [0] \cup [1] \cup [2] \cup [3].$$

**Exercise 9.19.** Where is the mistake?

**Theorem.** Let $S \neq \emptyset$ and let $R$ be a relation on $S$. If $R$ is symmetric and transitive, then it is reflexive.

*Proof.* Let $a \in S$. Consider any $b \in S$ such that $aRb$. By symmetry, $bRa$. By transitivity, $aRa$ and therefore, $R$ is reflexive. ■

# 10 Partial Order

## 1 Partially ordered sets

**Definition 10.1.** A relation $R$ on $S$ is a *partial order* if it is reflexive, antisymmetric, and transitive. The pair $(S, R)$ is called a *partially ordered set*, or *poset*.

Examples of partial orders

1. **Standard inequality on real numbers**
   Let $S = \mathbb{R}$. Define $xRy \iff x \leqslant y$.
2. **Subset inclusion on power sets**
   Let $A$ be a set and let $S = \mathcal{P}(A)$ be its power set. Define $xRy \iff x \subset y$.
   For instance, consider $A = \{1, 2, 3, 4\}$. Then
   - For $x = \{1, 2\}$ and $y = \{2\}$, we have $y \subset x$, so $yRx$.
   - For $x = \{1, 2\}$ and $y = \{2, 3\}$, we have $y \not\subset x$ ($y \not R x$) and $x \not\subset y$ ($x \not R y$).
3. **Divisibility relation on natural numbers**
   Let $S = \mathbb{N}$. Define $xRy \iff x \mid y$.
4. **Lexicographical order on words**
   Let $S$ be a set of words. Define $xRy$ if $x = y$ or $x$ precedes $y$ in dictionary order.
   *Example:* 'car' $R$ 'cat' because 'r' comes before 't' in the alphabet.

**Notation:** A partial order is commonly denoted by $\preccurlyeq$ or $\leqslant$. Thus, $(S, \preccurlyeq)$ is a poset. We will use this notation from now on.

A partial order need not compare every pair of elements. That is why it is called partial.

**Definition 10.2.** A partial order is a *total order* if for every $a, b \in S$,

$$a \preccurlyeq b \qquad \text{or} \qquad b \preccurlyeq a.$$

The ordinary order on $\mathbb{R}$ and lexicographic order are total. Set inclusion and divisibility generally are not.

**Exercise 10.3.** Let $S = \{1, 2, 3, 4\}$. Is there a relation on $S$ that is a partial order and an equivalence relation at the same time?

## 2 Extremal elements

For a poset $(S, \preccurlyeq)$, we define:

$$a \prec b \iff a \neq b \wedge a \preccurlyeq b$$

$$a \succcurlyeq b \iff b \preccurlyeq a$$

**Definition 10.4.** Let $(S, \preccurlyeq)$ be a poset.

- An element $g \in S$ is called a *greatest* element of $S$ if

$$a \preccurlyeq g \quad \forall a \in S$$

- An element $l \in S$ is called a *least* (or *smallest*) element of $S$ if

$$l \preccurlyeq a \quad \forall a \in S$$

- An element $m \in S$ is called a *maximal* element of $S$ if there is no element strictly greater than $m$, i.e.,

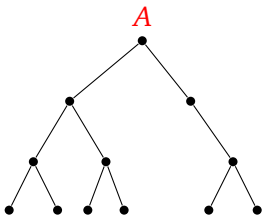
$$\neg(\exists a \in S : m \prec a)$$

- An element $p \in S$ is called a *minimal* element of $S$ if

$$\neg(\exists a \in S : a \prec p)$$

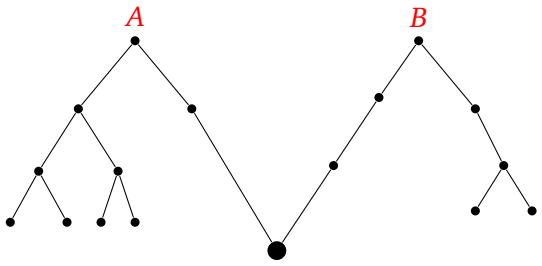
Examples

1. Let $S = [0, 1] \subset \mathbb{R}$ with the standard ordering $\preccurlyeq = \leqslant$.
   - **Greatest:** 1
   - **Maximal:** 1
   - **Least:** 0
   - **Minimal:** 0
2. Let $S = \{\{1\}, \{1, 2\}, \{1, 2, 3\}\}$ ordered by subset inclusion $x \preccurlyeq y \iff x \subset y$.
   - **Greatest:** $\{1, 2, 3\}$
   - **Maximal:** $\{1, 2, 3\}$
   - **Least:** $\{1\}$
   - **Minimal:** $\{1\}$
3. Let $S = \{\{1\}, \{1, 2\}, \{2, 3\}, \{2\}\}$ ordered by subset inclusion $x \preccurlyeq y \iff x \subset y$.
   - **Greatest:** None
   - **Maximal:** $\{1, 2\}$ and $\{2, 3\}$
   - **Least:** None
   - **Minimal:** $\{1\}$ and $\{2\}$
4. Let $S = \{\{1\}, \{1, 2\}, \{2, 3\}, \{2\}, \{4\}\}$ ordered by subset inclusion $x \preccurlyeq y \iff x \subset y$.
   - **Greatest:** None
   - **Maximal:** $\{1, 2\}$, $\{2, 3\}$, and $\{4\}$
   - **Least:** None
   - **Minimal:** $\{1\}$, $\{2\}$, and $\{4\}$
5. Let $S$ be the set of nodes of a rooted tree, where $x \preccurlyeq y$ if $x = y$ or $x$ is a descendant of $y$.




- **Greatest:** $A$ (the root)
- **Maximal:** $A$
- **Least:** None
- **Minimal:** Leaf nodes (the last nodes in each branch)

6. Let $S$ be a set of two trees with a shared leaf ordered by the descendant relation as before.




- **Greatest:** None (neither $A$ nor $B$ is comparable to the other)
- **Maximal:** $A$ and $B$
- **Least:** None (no single element is a descendant of all elements across both trees)
- **Minimal:** Leaf nodes

**Theorem 10.5.** *A poset has at most one greatest element.*

*Proof.* Suppose, for contradiction, that there are two distinct greatest elements $g_1, g_2 \in S$. Then:

$$g_1 \preccurlyeq g_2 \quad \text{and} \quad g_2 \preccurlyeq g_1.$$

By antisymmetry, $g_1 = g_2$, which yields a contradiction. ∎

**Theorem 10.6.** *The greatest element of a poset is its only maximal element.*

*Proof.* We break the proof into two parts:

1. The greatest element is maximal.
2. There are no other maximal elements.

**Part 1:** Let $g \in S$ be a greatest element, so $a \preccurlyeq g$ for all $a \in S$. If $g$ were not maximal, there would exist some $b \in S$ such that $g \prec b$, which contradicts $b \preccurlyeq g$. Thus, $g$ is maximal.

**Part 2:** Suppose there is another element $m \neq g$ that is also maximal. Since $g$ is the greatest element, $m \preccurlyeq g$. Because $m \neq g$, this implies $m \prec g$, which directly contradicts the maximality of $m$.

Hence, $g$ is the unique maximal element of $S$. ■

## 3 Bounds, supremum, and infimum

Let $(S, \preccurlyeq)$ be a poset and $A \subset S$. Since $(A, \preccurlyeq)$ is also a poset, we can define its greatest/least and maximal/minimal elements in the same way as they were defined for $S$. A more interesting is the following definition.

**Definition 10.7.** Let $(S, \preccurlyeq)$ be a poset and $A \subset S$.

- An element $u \in S$ is called an *upper bound* of $A$ if

$$a \preccurlyeq u \quad \forall a \in A$$

  The least upper bound of $A$ is called the *supremum* of $A$, denoted $\sup A$.

- An element $l \in S$ is called a *lower bound* of $A$ if

$$l \preccurlyeq a \quad \forall a \in A$$

  The greatest lower bound of $A$ is called the *infimum* of $A$, denoted $\inf A$.

**Exercise 10.8.** Prove that the supremum of a subset $A$ of a poset $(P, \preccurlyeq)$, if it exists, is unique. Prove the analogous statement for the infimum.

Examples

In the following examples, greatest, maximal, least, and minimal elements are always taken in $A$. Upper and lower bounds are taken in $S$.

1. Let $S = [0, 2] \subset \mathbb{R}$ with the standard order $\preccurlyeq = \leqslant$ and $A = [0, 1]$.
   - **Greatest:** 1
   - **Maximal:** 1
   - **Least:** 0
   - **Minimal:** 0
   - **Upper bounds of** $A$**:** $[1, 2]$
   - **Lower bounds of** $A$**:** $\{0\}$
   - $\sup A = 1$
   - $\inf A = 0$
2. Let $S = [0, 2] \subset \mathbb{R}$ with the standard order $\preccurlyeq = \leqslant$ and $A = (0, 1)$.
   - **Greatest:** None
   - **Maximal:** None
   - **Least:** None
   - **Minimal:** None

- **Upper bounds of** $A$**:** $[1, 2]$
- **Lower bounds of** $A$**:** $\{0\}$
- $\sup A = 1$
- $\inf A = 0$

Neither 0 nor 1 belongs to $A$, although both are bounds in $S$.

3. Let $S = [0, 2] \subset \mathbb{R}$ with the standard order $\preccurlyeq = \leqslant$ and $A = [0, 1)$.
   - **Greatest:** None
   - **Maximal:** None
   - **Least:** 0
   - **Minimal:** 0
   - **Upper bounds of** $A$**:** $[1, 2]$
   - **Lower bounds of** $A$**:** $\{0\}$
   - $\sup A = 1$
   - $\inf A = 0$
4. Let $S = [0, 2] \subset \mathbb{R}$ with the standard order $\preccurlyeq = \leqslant$ and $A = (0, 1]$.
   - **Greatest:** 1
   - **Maximal:** 1
   - **Least:** None
   - **Minimal:** None
   - **Upper bounds of** $A$**:** $[1, 2]$
   - **Lower bounds of** $A$**:** $\{0\}$
   - $\sup A = 1$
   - $\inf A = 0$
5. Let $S = \{\{1\}, \{2\}, \{1, 2\}, \{2, 3\}\}$, ordered by inclusion and $A = \{\{1\}, \{2\}\}$.
   - **Greatest:** None
   - **Maximal:** $\{1\}$ and $\{2\}$
   - **Least:** None
   - **Minimal:** $\{1\}$ and $\{2\}$
   - **Upper bounds of** $A$**:** $\{\{1, 2\}\}$
   - **Lower bounds of** $A$**:** None
   - $\sup A = \{1, 2\}$
   - $\inf A$ does not exist

   The supremum belongs to $S$, but not to $A$.
6. Let $S = \{1, 2, 3, 6, 12\}$, ordered by divisibility, and $A = \{2, 3\}$.
   - **Greatest:** None

- **Maximal:** 2 and 3
- **Least:** None
- **Minimal:** 2 and 3
- **Upper bounds of** $A$**:** $\{6, 12\}$
- **Lower bounds of** $A$**:** $\{1\}$
- $\sup A = 6$
- $\inf A = 1$

7. Keep the same divisibility poset $S$, but let $A = \{2, 3, 6\}$.

   - **Greatest:** 6
   - **Maximal:** 6
   - **Least:** None
   - **Minimal:** 2 and 3
   - **Upper bounds of** $A$**:** $\{6, 12\}$
   - **Lower bounds of** $A$**:** $\{1\}$
   - $\sup A = 6$
   - $\inf A = 1$

8. Let $S = \{0, 1, 2\}^2$ with the order $(a, b) \preccurlyeq (c, d)$ if $a \leqslant c$ and $b \leqslant d$. Let $A = \{(0, 2), (1, 1), (2, 0)\}$.

   - **Greatest:** None
   - **Maximal:** $(0, 2)$, $(1, 1)$, and $(2, 0)$
   - **Least:** None
   - **Minimal:** $(0, 2)$, $(1, 1)$, and $(2, 0)$
   - **Upper bounds of** $A$**:** $\{(2, 2)\}$
   - **Lower bounds of** $A$**:** $\{(0, 0)\}$
   - $\sup A = (2, 2)$
   - $\inf A = (0, 0)$

   The three elements of $A$ are pairwise incomparable.

9. Let $S = \{\varnothing, \{1\}, \{2\}, \{1, 2, 3\}, \{1, 2, 4\}\}$, ordered by inclusion, and let $A = \{\{1\}, \{2\}\}$.

   - **Greatest:** None
   - **Maximal:** $\{1\}$ and $\{2\}$
   - **Least:** None
   - **Minimal:** $\{1\}$ and $\{2\}$
   - **Upper bounds of** $A$**:** $\{\{1, 2, 3\}, \{1, 2, 4\}\}$
   - **Lower bounds of** $A$**:** $\{\varnothing\}$
   - $\sup A$ does not exist
   - $\inf A = \varnothing$

The two upper bounds are incomparable, so there is no least upper bound.

10. Let $S = \{\varnothing, \{1\}, \{2\}, \{1,2,3\}, \{1,2,4\}, \{1,2,3,4\}\}$, ordered by inclusion, and let $A = \{\{1,2,3\}, \{1,2,4\}\}$.

    - **Greatest:** None
    - **Maximal:** $\{1,2,3\}$ and $\{1,2,4\}$
    - **Least:** None
    - **Minimal:** $\{1,2,3\}$ and $\{1,2,4\}$
    - **Upper bounds of** $A$**:** $\{\{1,2,3,4\}\}$
    - **Lower bounds of** $A$**:** $\{\varnothing, \{1\}, \{2\}\}$
    - $\sup A = \{1,2,3,4\}$
    - $\inf A$ does not exist

    The lower bounds $\{1\}$ and $\{2\}$ are incomparable, so there is no greatest lower bound.

11. Let $S = \mathbb{R}$ with its usual order and $A = \{(-1)^n(1 - 1/n) : n \in \mathbb{N},\ n \geqslant 1\}$.

    - **Greatest:** None
    - **Maximal:** None
    - **Least:** None
    - **Minimal:** None
    - **Upper bounds of** $A$**:** $[1, \infty)$
    - **Lower bounds of** $A$**:** $(-\infty, -1]$
    - $\sup A = 1$
    - $\inf A = -1$

    For even $n$, the values $1 - 1/n$ approach 1 from below; for odd $n$, the values $-1 + 1/n$ approach $-1$ from above. Neither limit belongs to $A$.

12. Let $S = \mathbb{R}$ with its usual order and $A = \{\sqrt{n+1} - \sqrt{n} : n \in \mathbb{N},\ n \geqslant 1\}$.

    - **Greatest:** $\sqrt{2} - 1$
    - **Maximal:** $\sqrt{2} - 1$
    - **Least:** None
    - **Minimal:** None
    - **Upper bounds of** $A$**:** $[\sqrt{2} - 1, \infty)$
    - **Lower bounds of** $A$**:** $(-\infty, 0]$
    - $\sup A = \sqrt{2} - 1$
    - $\inf A = 0$

    *Hint:* Use $\sqrt{n+1} - \sqrt{n} = 1/(\sqrt{n+1} + \sqrt{n})$.

13. Let $S = \mathbb{R}$ with its usual order and

$$A = \left\{ \frac{m}{m+n} : m, n \in \mathbb{N},\ 1 \leqslant m < n \right\}.$$

- **Greatest:** None
- **Maximal:** None
- **Least:** None
- **Minimal:** None
- **Upper bounds of** $A$**:** $[1/2, \infty)$
- **Lower bounds of** $A$**:** $(-\infty, 0]$
- $\sup A = 1/2$
- $\inf A = 0$

Since $m < n$, every value is less than $1/2$. Taking $m = n - 1$ and letting $n$ grow approaches $1/2$; taking $m = 1$ and letting $n$ grow approaches 0.

14. Let $S = \mathbb{R}$ with its usual order and $A = \{xy : x, y > 0,\ x + y = 1\}$.

- **Greatest:** $1/4$
- **Maximal:** $1/4$
- **Least:** None
- **Minimal:** None
- **Upper bounds of** $A$**:** $[1/4, \infty)$
- **Lower bounds of** $A$**:** $(-\infty, 0]$
- $\sup A = 1/4$
- $\inf A = 0$

The inequality $(x-y)^2 \geqslant 0$ gives $4xy \leqslant (x+y)^2 = 1$, with equality at $x = y = 1/2$. Letting $x$ approach 0 makes $xy$ approach 0 without attaining it.

15. Let $S = \mathbb{R}$ with its usual order and $A = \{q \in \mathbb{Q} : q^2 < 2\}$.

- **Greatest:** None
- **Maximal:** None
- **Least:** None
- **Minimal:** None
- **Upper bounds of** $A$**:** $[\sqrt{2}, \infty)$
- **Lower bounds of** $A$**:** $(-\infty, -\sqrt{2}]$
- $\sup A = \sqrt{2}$
- $\inf A = -\sqrt{2}$

The condition $q^2 < 2$ means $-\sqrt{2} < q < \sqrt{2}$. By the density of rational numbers, elements of $A$ approach both endpoints, but neither endpoint is rational.

16. Let $S = \mathbb{R}$ with its usual order and

$$A = \left\{ \frac{n}{n+1} + \frac{(-1)^n}{n} : n \in \mathbb{N},\ n \geqslant 1 \right\}.$$

- **Greatest:** $7/6$

- **Maximal:** $7/6$
- **Least:** $-1/2$
- **Minimal:** $-1/2$
- **Upper bounds of** $A$**:** $[7/6, \infty)$
- **Lower bounds of** $A$**:** $(-\infty, -1/2]$
- $\sup A = 7/6$
- $\inf A = -1/2$

For even $n$, the expression is $1 + 1/(n(n+1))$, whose largest value occurs at $n = 2$. For odd $n$, it is $1 - 1/n - 1/(n+1)$, whose smallest value occurs at $n = 1$.

**Exercise 10.9.** Let $S$ be the set of all equivalence relations on $\mathbb{Z}$, ordered by inclusion. Define $R, T \in S$ by

$$aRb \iff a = b \text{ or } a + b = 0, \qquad aTb \iff a = b \text{ or } a + b = 6.$$

Determine $\inf\{R, T\}$ and $\sup\{R, T\}$ in $S$.

*Hint:* For the infimum, consider $R \cap T$. For the supremum, find the smallest equivalence relation containing both $R$ and $T$.

In some examples above we used the following "obvious"[1] fact.

**Proposition 10.10** (Completeness of $\mathbb{R}$)**.** *Every nonempty subset of* $\mathbb{R}$ *that is bounded above has a supremum in* $\mathbb{R}$*. Similarly, every nonempty subset of* $\mathbb{R}$ *that is bounded below has an infimum in* $\mathbb{R}$*.*

We omit the proof. Informally, completeness means that the real line has no gaps. For example, the rational numbers leave a gap at $\sqrt{2}$, whereas the real numbers contain this point.

[1] It is a highly nontrivial fact